

Unit 3

Module 3 : Ordinary Differential Equations of Higher Orders

3

Higher Order Linear Ordinary Differential Equations

Syllabus

Higher order linear differential equations with constant and variable coefficients, Euler-Cauchy equations, solution by variation of parameters, method of undetermined coefficients.

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3.1 Introduction

A differential equation of the form $F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}\right) = 0$ in which the dependent variable $y(x)$ and its derivatives viz. $\frac{dy}{dx}, \frac{d^2y}{dx^2}$ etc. occur in first degree and are not multiplied together is called a linear differential equation.

3.2 Linear Differential Equations (LDE) with Constant Coefficients

A general linear differential equation of n^{th} order with constant coefficients is given by :

$$k_0 \frac{d^ny}{dx^n} + k_1 \frac{d^{n-1}y}{dx^{n-1}} + \dots + k_{n-1} \frac{dy}{dx} + k_n y = F(x)$$

where k s are constant and $F(x)$ is a function of x alone or constant.

$$\Rightarrow (k_0 D^n + k_1 D^{n-1} + \dots + k_{n-1} D + k_n) y = F(x)$$

Or $f(D)y = F(x)$, where $D^n \equiv \frac{d^n}{dx^n}$, $D^{n-1} \equiv \frac{d^{n-1}}{dx^{n-1}}$, ..., $D \equiv \frac{d}{dx}$ are called differential operators.

3.3 Solving Linear Differential Equations with Constant Coefficients

GTU : Summer-18,20,23,24, Winter-18,19,22,23

Complete solution of equation $f(D)y = F(x)$ is given by $y = \text{C.F.} + \text{P.I.}$ where C.F. denotes complimentary function and P.I. is particular integral.

when $F(x) = 0$, then solution of equation $f(D)y = 0$ is given by $y = \text{C.F.}$

3.3.1 Rules for Finding Complimentary Function (C.F.)

Consider the equation $f(D)y = F(x)$

$$\Rightarrow (k_0 D^n + k_1 D^{n-1} + \dots + k_{n-1} D + k_n) y = F(x)$$

Step 1 : Put $D = m$, auxiliary equation (A.E.) is given by $f(m) = 0$

$$\Rightarrow k_0 m^n + k_1 m^{n-1} + \dots + k_{n-1} m + k_n = 0$$

...(3.3.1)

Step 2 : Solve the auxiliary equation given by (3.3.1)

I. If the n roots of A.E. are real and distinct say $m_1, m_2, \dots, m_n$

$$\text{C.E.} = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_n e^{m_n x}$$

II. If two or more roots are equal i.e. $m_1 = m_2 = \dots = m_k, k \leq n$

$$\text{C.F.} = (C_1 + C_2 x + C_3 x^2 + \dots + C_k x^{k-1}) e^{m_1 x} + \dots + C_n e^{m_n x}$$

III. If A.E. has a pair of imaginary roots i.e.

$$m_1 = \alpha + i\beta, m_2 = \alpha - i\beta$$

$$\text{C.F.} = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x) + C_3 e^{m_3 x} + \dots + C_n e^{m_n x}$$

IV. If 2 pairs of imaginary roots are equal i.e.

$$m_1 = m_2 = \alpha + i\beta$$

$$m_3 = m_4 = \alpha - i\beta$$

$$\text{C.F.} = e^{\alpha x} [(C_1 + C_2 x) \cos \beta x + (C_3 + C_4 x) \sin \beta x] + \dots + C_n e^{m_n x}$$

Ex. 3.3.1 : Solve the differential equation :

$$\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 0.$$

Sol. : $\Rightarrow (D^2 - 8D + 15)y = 0$

Auxiliary equation is : $m^2 - 8m + 15 = 0$

$$\Rightarrow (m-3)(m-5) = 0$$

$$\Rightarrow m = 3, 5$$

$$\text{C.F.} = C_1e^{3x} + C_2e^{5x}$$

Since $F(x) = 0$, solution is given by $y = \text{C.F.}$

$$\Rightarrow y = C_1e^{3x} + C_2e^{5x}$$

Ex. 3.3.2 : Solve the differential equation :

$$\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0.$$

Sol. : $\Rightarrow (D^3 - 6D^2 + 11D - 6)y = 0$

Auxiliary equation is :

$$m^3 - 6m^2 + 11m - 6 = 0 \quad \dots (1)$$

By hit and trial $(m-2)$ is a factor of (1)

$\therefore$ (1) may be rewritten as

$$m^3 - 2m^2 - 4m^2 + 8m + 3m - 6 = 0$$

$$\Rightarrow m^2(m-2) - 4m(m-2) + 3(m-2) = 0$$

$$\Rightarrow (m^2 - 4m + 3)(m-2) = 0$$

$$\Rightarrow (m-3)(m-1)(m-2) = 0$$

$$\Rightarrow m = 1, 2, 3$$

$$\text{C.F.} = C_1e^x + C_2e^{2x} + C_3e^{3x}$$

Since $F(x) = 0$, solution is given by $y = \text{C.F.}$

$$\Rightarrow y = C_1e^x + C_2e^{2x} + C_3e^{3x}$$

Ex. 3.3.3 : Solve $(D^4 - 10D^3 + 35D^2 - 50D + 24)y = 0$.

Sol. : Auxiliary equation is :

$$m^4 - 10m^3 + 35m^2 - 50m + 24 = 0 \quad \dots (1)$$

By hit and trial $(m-1)$ is a factor of (1)

$\therefore$ (1) may be rewritten as

$$m^4 - m^3 - 9m^3 + 9m^2 + 26m^2 - 26m - 24m + 24 = 0$$

$$\Rightarrow m^3(m-1) - 9m^2(m-1) + 26m(m-1) - 24(m-1) = 0$$

$$\Rightarrow (m-1)(m^3 - 9m^2 + 26m - 24) = 0 \quad \dots (2)$$

By hit and trial $(m-2)$ is a factor of (2)

$\therefore$ (2) may be rewritten as

$$(m-1)(m^3 - 2m^2 - 7m^2 + 14m + 12m - 24) = 0$$

$$\Rightarrow (m-1)[m^2(m-2) - 7m(m-2) + 12(m-2)] = 0$$

$$\Rightarrow (m-1)(m^2 - 7m + 12)(m-2) = 0$$

$$\Rightarrow (m-1)(m-3)(m-4)(m-2) = 0$$

$$\Rightarrow m = 1, 2, 3, 4$$

$$\text{C.F.} = C_1e^x + C_2e^{2x} + C_3e^{3x} + C_4e^{4x}$$

Since $F(x) = 0$, solution is given by $y = \text{C.F.}$

$$\Rightarrow y = C_1e^x + C_2e^{2x} + C_3e^{3x} + C_4e^{4x}$$

Ex. 3.3.4 : Solve the differential equation :

$$\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0.$$

Sol. : $\Rightarrow (D^3 + 2D^2 + D)y = 0$

Auxiliary equation is : $m^3 + 2m^2 + m = 0$

$$\Rightarrow m(m^2 + 2m + 1) = 0$$

$$\Rightarrow m(m+1)^2 = 0$$

$$\Rightarrow m = 0, -1, -1$$

$$\text{C.F.} = C_1 + (C_2 + C_3x)e^{-x}$$

Since $F(x) = 0$, solution is given by $y = \text{C.F.}$

$$\Rightarrow y = C_1 + (C_2 + C_3x)e^{-x}$$

Ex. 3.3.5 : Solve the differential equation :

$$\frac{d^4y}{dx^4} - 2\frac{d^2y}{dx^2} + y = 0.$$

Sol. : $\Rightarrow (D^4 - 2D^2 + 1)y = 0$

Auxiliary equation is : $m^4 - 2m^2 + 1 = 0$

$$\Rightarrow m(m^2 - 1)^2 = 0$$

$$\Rightarrow (m+1)^2(m-1)^2 = 0$$

$$\Rightarrow m = -1, -1, 1, 1$$

$$\text{C.F.} = (C_1 + C_2x)e^{-x} + (C_3 + C_4x)e^x$$

Since $F(x) = 0$, solution is given by $y = \text{C.F.}$

$$\Rightarrow y = (C_1 + C_2 x)e^{-x} + (C_3 + C_4 x)e^x$$

Ex. 3.3.6 : Solve the differential equation : $\frac{d^3 y}{dx^3} - 2\frac{dy}{dx} + 4y = 0$.

Sol. : $\Rightarrow (D^3 - 2D^2 + 4)y = 0$

Auxiliary equation is : $m^3 - 2m + 4 = 0$... (1)

By hit and trial $(m + 2)$ is a factor of (1)

$\therefore$ (1) may be rewritten as

$$m^3 + 2m^3 - 2m^2 - 4m + 2m + 4 = 0$$

$$\Rightarrow m^2(m+2) - 2m(m+2) + 2(m+2) = 0$$

$$\Rightarrow (m+2)(m^2 - 2m + 2) = 0$$

$$\Rightarrow m = -2, 1 \pm i$$

$$\text{C.F.} = C_1 e^{-2x} + e^x (C_2 \cos x + C_3 \sin x)$$

Since $F(x) = 0$, solution is given by $y = \text{C.F.}$

$$\Rightarrow y = C_1 e^{-2x} + e^x (C_2 \cos x + C_3 \sin x)$$

Ex. 3.3.7 : Solve the differential equation : $(D^2 - 2D + 5)^2 y = 0$.

Sol. :

Auxiliary equation is : $(m^2 - 2m + 5)^2$... (1)

Solving (1), we get

$$\Rightarrow m = 1 \pm 2i, 1 \pm 2i$$

$$\text{C.F.} = e^x [(C_1 + C_2 x) \cos 2x + (C_3 + C_4 x) \sin 2x]$$

Since $F(x) = 0$, solution is given by $y = \text{C.F.}$

$$\Rightarrow y = e^x [(C_1 + C_2 x) \cos 2x + (C_3 + C_4 x) \sin 2x]$$

Ex. 3.3.8 : Solve the differential equation : $(D^2 + 4)^3 y = 0$.

Sol. : Auxiliary equation is : $(m^2 + 4)^3$... (1)

Solving (1), we get

$$\Rightarrow m = \pm 2i, \pm 2i, \pm 2i$$

$$\text{C.F.} = (C_1 + C_2 x + C_3 x^2) \cos 2x + (C_4 + C_5 x + C_6 x^2) \sin 2x$$

Since $F(x) = 0$, solution is given by $y = \text{C.F.}$

$$\Rightarrow y = (C_1 + C_2 x + C_3 x^2) \cos 2x + (C_4 + C_5 x + C_6 x^2) \sin 2x$$

Ex. 3.3.9 : Solve $y'' - 4y' + 4y = 0$

GTU : Winter-22,19, AEM, Marks 3

Sol. :

Here : $y'' - 4y' + 4y = 0$

∴ A.E. $D^2 - 4D + 4 = 0$

$$(D-2)^2 = 0$$

$$D = 2, 2$$

$$y = (C_1 + C_2 x) e^{2x}$$

Ex. 3.3.10 : Solve $y''' - 6y'' + 11y' - 6y = 0$

GTU : Summer-18, AEM, Marks 3

Sol. : Here, $y''' - 6y'' + 11y' - 6y = 0$ is given

A.E. $D^3 - 6D^2 + 11D - 6 = 0$

$$∴ D^3 - D^2 - 5D^2 + 5D + 6D - 6 = 0$$

$$∴ D^2(D-1) - 5D(D-1) + 6(D-1) = 0$$

$$∴ (D-1)(D^2 - 5D + 6) = 0$$

$$∴ (D-1)(D-2)(D-3) = 0$$

$$∴ D = 1, 2, 3$$

$$∴ y = C_1 e^x + C_2 e^{2x} + C_3 e^{3x}$$

Ex. 3.3.11 : Solve $\frac{d^3 y}{dx^3} - 3\frac{dy}{dx} + 2y = 0$

GTU : Summer-20, AEM, Marks 3

Sol. : Here $y''' - 3y' + 2y = 0$ is given

A.E. $D^3 - 3D^2 + 2 = 0$

$$∴ D^3 - D^2 + D^2 - D - 2D + 2 = 0$$

$$∴ D^2(D-1) + D(D-1) - 2(D-1) = 0$$

$$∴ (D-1)(D^2 + D - 2) = 0$$

$$∴ (D-1)(D^2 + 2D - D - 2) = 0$$

$$∴ (D-1)(D(D+2) - 1(D+2)) = 0$$

$$∴ (D-1)(D-1)(D+2) = 0$$

$$∴ D = 1, 1, -2$$

$$∴ y = e^x + (C_1 + C_2 x) + C_3 e^{-2x}$$

Ex. 3.3.12 : Solve $y'' + 3y' + 2y = 0$

GTU : Summer-24, AEM, Marks 3

Sol. :

Here $y'' + 3y' + 2y = 0$ is given

$$∴ D^2 + 3D + 2 = 0$$

$$D^2 + 2D + D + 2 = 0$$

$$∴ D(D+2) + 1(D+2) = 0$$

$$∴ D + 1 = 0, D + 2 = 0$$

$$∴ D = -1 \text{ and } D = -2$$

$$∴ y = C_1 e^{-x} + C_2 e^{-2x}$$

Ex. 3.3.13 Solve $\frac{d^4 y}{dx^4} - 2\frac{d^2 y}{dx^2} + y = 0$

GTU : Winter-18, AEM, Marks 3

Sol. :

Here $y^{iv} - 2y'' + y = 0$

A.E. $D^4 - 2D^2 + 1 = 0$

$$∴ (D^2 - 1)^2 = 0$$

$$∴ D^2 = 1 \text{ two times}$$

$$∴ D = \pm 1 \text{ two times}$$

$$∴ y = e^x (C_1 + C_2 x) + e^{-x} (C_3 + C_4 x)$$

Ex. 3.3.14 : Solve $y'' + 6y' + 9y = 0$, $y(0) = 1$, $y'(0) = 2$

GTU : Summer-23, Maths-2, Marks 4

Sol. : Here $y'' + 6y' + 9y = 0$

A.E. $D^2 + 6D + 9 = 0$

$$(D+3)^2 = 0$$

$$D = -3, -3$$

$$∴ y = e^{-3x} (C_1 + C_2 x)$$

Here, $y(0) = 1 \Rightarrow y = 1$ When $x = 0$

$$1 = C_1$$

$$y^1 = -3e^{-3x} (C_1 + C_2 x) + e^{-3x} C_2$$

Here, $y'(0) = 2 \Rightarrow y' = 2$ when $x = 0$

$$2 = -3C_1 + C_2 \Rightarrow 2 = -3 + C_1 \Rightarrow C_2 = 5$$

$$∴ y = e^{-3x} (1 + 5x)$$

Ex. 3.3.15 : Solve $y'' - 5y' - 14y = 0$, $y(0) = 3$, $y'(0) = 1$.

GTU : Summer-23, Maths-2, Marks 4

Sol. : Here $y'' - 5y' - 14y = 0$ is given

A.E. $D^2 - 5D - 14 = 0$

$$D^2 - 7D + 2D - 14 = 0$$

$$D(D - 7) + 2(D - 7) = 0$$

$$(D - 7)(D + 2) = 0$$

$\therefore D = 7, D = -2$

$$y = C_1 e^{7x} + C_2 e^{-2x}$$

Here, $y(0) = 3 \Rightarrow y = 3$ when $x = 0$

$$3 = C_1 + C_2 \quad \dots(1)$$

Here, $y'(0) = 1 \Rightarrow y' = 1$ when $x = 0$

$$y' = 7C_1 e^{7x} - 2C_2 e^{-2x}$$

$$1 = 7C_1 - 2C_2 \quad \dots(2)$$

$$\begin{array}{l} \therefore C_1 + C_2 = 3 \\ 7C_1 - 2C_2 = 1 \end{array} \quad \begin{array}{l} 2C_1 + 2C_2 = 6 \\ 7C_1 - 2C_2 = 1 \end{array}$$

$$9C_1 = 7 \Rightarrow C_1 = 7/9$$

and $7/9 + C_2 = 3 \Rightarrow C_2 = 3 - 7/9 = \frac{27-7}{9} = \frac{20}{9}$

$\therefore y = 7/9 e^{7x} + 20/9 e^{-2x}$

Ex. 3.3.16 : Solve i) $(D^2 + 7D - 18)y = 0$

ii) $y'' + y' + 2y = 0$

GTU : Winter-23, Maths-2, Marks 4

Sol. : i) $(D^2 + 7D - 18)y = 0$

A.E. $D^2 + 7D - 18 = 0$

$$D^2 + 9D - 2D - 18 = 0$$

$$D(D + 9) - 2(D + 9) = 0$$

$$(D - 2)(D + 9) = 0$$

$\therefore D = 2$ and $D = -9$

$\therefore y = C_1 e^{2x} + C_2 e^{-9x}$

ii) $y'' + y' + 2y = 0$

A.E. $D^2 + D + 2 = 0$

$a = 1, b = 1, c = ?$

$$\Delta = b^2 - 4ac = 1 - 8 = -7$$

$$D_1, D_2 = \frac{-b \pm \sqrt{\Delta}}{2a}$$

$$= \frac{-1 \pm \sqrt{-7}}{2} = \frac{-1}{2} \pm \frac{\sqrt{7}}{2} i$$

$$y = e^{-1/2x} \left(C_1 \cos \frac{\sqrt{7}}{2} x + C_2 \sin \frac{\sqrt{7}}{2} x \right)$$

Ex. 3.3.17 : Solve $2D^2 y + Dy - 6y = 0$.

GTU : Winter-22, Maths-2, Marks 3

Sol. : Here, $(2D^2 + D - 6)y = 0$

A.E. $2D^2 + D - 6 = 0$

$\therefore 2D^2 + 4D - 3D - 6 = 0$

$\therefore 2D(D + 2) - 3(D + 2) = 0$

$$(2D - 3)(D + 2) = 0$$

$$2D = 3, D + 2 = 0$$

$\Rightarrow D = 3/2, D = -2$

$\therefore y = C_1 e^{3/2x} + C_2 e^{-2x}$

3.4 Method of Undetermined Coefficients

**GTU : Summer-18,19,20,21,22,23,24
Winter-18,19,21,20,21,22,23**

Here we discuss technique for finding the particular integral of the linear differential equation with constant co-efficient.

$$\frac{dy^n}{dx^n} + C_1 \frac{dy^{n-1}}{dx^{n-1}} + C_2 \frac{dy^{n-2}}{dx^{n-2}} + \dots + a_n y = r(x) \quad \dots (3.4.1)$$

We know that the general solution of (3.4.1) can be written as $y(x) = y_C(x) + y_P(x)$.

where $y_C(x)$ is the complementary function of the homogenous part of equation (3.4.1).

To find the particular integral $y_P(x)$ of (3.4.1), we will use this method whenever the non-homogenous term is elementary function like, polynomial term (x^n), exponential term (e^{ax}) sine or cosine term ($\sin ax$ or $\cos ax$), finite sum and/or product of these function.

Here, first we assume trail solution y_P depending on (R.H.S. term) or $r(x)$ or non-homogenous terms as per the following Table. 3.4.1.

Sr. No.	R.H.S. at equations (3.4.1) Term or $r(x)$	Trail solution y_P
1.	$r(x) = e^{ax}$	$y_P = C \cdot e^{ax}$
2.	$r(x) = \sin ax$ or $\cos ax$ or $\sin ax + \cos ax$	$y_P = A \cos ax + B \sin ax$
3.	$r(x) = C_1 + C_2x + C_3x^2 + \dots C_nx^n$ Example $r(x) = a + bx + cx^2 + dx^3$ $r(x) = a + bx + cx^2$ $r(x) = a + bx$ $r(x) = a$	$y_P = Ax^n + Bx^{n-1} + Cx^{n-2} + \dots$ $y_P = Ax^3 + Bx^2 + Cx + D$ $y_P = Ax^2 + Bx + C$ $y_P = Ax + B$ $y_P = A$
4.	$r(x) = e^{ax} \sin bx$ or $e^{ax} \cos bx$	$y_P = e^{ax}(A \cos bx + B \sin bx)$
5.	$r(x) = xe^{ax}$ $r(x) = x^2 e^{ax}$	$y_P = e^{ax}(Ax + B)$ $y_P = e^{ax}(Ax^2 + Bx + C)$
6.	$r(x) = x \sin ax$ $r(x) = x^2 \sin ax$	$y_P = (Ax + B) \cos ax + (Cx + D) \sin ax$ $y_P = (Ax^2 + Bx + C) \cos ax + (Dx^2 + Ex + F) \sin ax$

Table 3.4.1

Additional rules for assuming trail solution y_P .

- If no term of $r(x)$ occurs in C.F. or y_C then suitable P.I. assumed from the above table.
- If any term of $r(x)$ is also present in C.F. or y_C only once then multiple x corresponding to y_P . If any term of $r(x)$ is also present in C.F. or y_C twice then multiply with x^2 corresponding to y_P term and so on.
- If $r(x) = \sin ax$ or $\cos ax$ or $\cos ax$ or addition of $\sin ax + \cos ax$ then all term have same y_P .

Some example to assume trail solution y_P .

$$\text{i) If } r(x) = \begin{Bmatrix} e^{3x} \\ 2e^{3x} \\ 4e^{3x} \\ C \cdot e^{3x} \end{Bmatrix} \rightarrow y_P = C \cdot e^{3x} \quad r(x) = \begin{Bmatrix} e^{bx} \\ 5e^{bx} \\ 6e^{bx} \\ C \cdot e^{bx} \end{Bmatrix} \rightarrow y_P = C \cdot e^{bx}$$

$$\text{ii) } r(x) = \begin{Bmatrix} \sin 2x \\ \cos 2x \\ 2 \sin 2x \\ 3 \cos 2x \\ \sin 2x + \cos 2x \end{Bmatrix} \rightarrow y_P = A \cos 2x + B \sin 2x$$

$$\text{iii) } r(x) = \begin{Bmatrix} ax^3 + bx^2 \\ 2x^3 + 3x^2 + x \\ x^3 + x + 1 \\ x^3 + x^2 + x + 1 \end{Bmatrix} \rightarrow y_P = Ax^3 + Bx^2 + Cx + D$$

any 3rd degree polynomial

Ex. 3.4.1 : Solve the differential equation $(D^2 + 4)y = x^2 + \sin 2x$.

Sol. : Here, $(D^2 + 4)y = x^2 + \sin 2x$ is given

∴ For C.F.

$$D^2 + 4 = 0$$

∴

$$D^2 = -4 \Rightarrow D = \pm 2i$$

Complex roots.

∴

$$\text{C.F.} = y_C = C_1 \cos 2x + C_2 \sin 2x$$

Here, we use undeterminant co-efficient method to find P.I. (y_P)

∴ Let, the trial solution are,

$$y_P = Ax^2 + Bx + C + D \cos 2x + E \sin 2x$$

Here $\sin 2x$ term appears once in y_C

∴ We multiply the term corresponding to $\sin 2x$ with x .

∴

$$y_P = Ax^2 + Bx + C + Dx \cos 2x + Ex \sin 2x$$

∴

$$y'_P = 2Ax + B + D \cos 2x - 2Dx \sin 2x + E \sin 2x + 2Ex \cos 2x$$

$$y''_P = 2A - 2D \sin 2x - 2D \sin 2x - 4Dx \cos 2x + 2E \cos 2x + 2E \cos 2x - 4Ex \sin 2x$$

$$y''_P = 2A - 4D \sin 2x - 4Dx \cos 2x - 4E \cos 2x - 4Ex \sin 2x$$

Substituting y'_P and y''_P in given equation.

$$y''_P + 4y_P = x^2 + \sin 2x$$

$$2A - 4D \sin 2x - 4Dx \cos 2x + 4E \cos 2x - 4Ex \sin 2x + 4Ax^2 + 4Bx + 4C + 4Dx \cos 2x + 4Ex \sin 2x = x^2 + \sin 2x$$

$$\therefore 4Ax^2 + 4Bx + (2A + 4C) - 4D \sin 2x + 4E \cos 2x = x^2 + \sin 2x$$

Comparing the corresponding co-efficient from both sides,

$$4A = 1 \Rightarrow A = \frac{1}{4}$$

$$4B = 0 \Rightarrow B = 0$$

$$2A + 4C = 0 \Rightarrow 2 \times \frac{1}{4} + 4C = 0$$

$$\Rightarrow 4C = -\frac{1}{2}$$

$$\Rightarrow C = -\frac{1}{8}$$

$$-4D = 1 \Rightarrow D = -\frac{1}{4}$$

$$4E = 0 \Rightarrow E = 0$$

Hence, P.I.

$$y_P = \frac{1}{4}x^2 - \frac{1}{8} - \frac{1}{4}x \cdot \cos 2x$$

∴

$$y = y_C + y_P$$

$$= C_1 \cos 2x + C_2 \sin 2x + \frac{1}{4}x^2 - \frac{1}{8} - \frac{1}{4}x \cdot \cos 2x$$

Ex. 3.4.2 : Using method of undeterminant co-efficient solve $y'' - 2y' + y = e^x + x$.

GTU : Summer-18, Maths-3, Marks 3

Sol. : Here, $y'' - 2y' + y = e^x + x$ is given

$\therefore$ For C.F. we write.

$$D^2 - 2D + 1 = 0$$

$$(D-1)^2 = 0$$

$$D = 1, 1 \text{ (two times)}$$

$$\text{C.F.} = e^x(C_1 + C_2x)$$

Now, for P.I. we use U.C. method

$\therefore$ Let us take, trail solution,

$$y_P = C_1 \cdot e^x + C_2x + C_3$$

Here, e^x term appears two times in C.F. so, we multiply the term corresponding to e^x with x^2

$$\therefore y_P = C_1x^2e^x + C_2x + C_3$$

$$\therefore y'_P = 2C_1xe^x + C_1x^2e^x + C_2$$

$$y''_P = 2C_1e^x + 2C_1xe^x + 2C_1xe^x + C_1x^2e^x$$

Now put y_P, y'_P and y''_P in given equation,

$$\therefore y''_P - 2y'_P + y_P = e^x + x$$

$$\therefore 2C_1e^x + 4C_1xe^x + C_1x^2e^x$$

$$- 4C_1xe^x - 2C_1x^2e^x - 2C_2 + C_1e^xx^2 + C_2x + C_3 = e^x + x$$

$$\therefore 2C_1e^x + C_2x + (C_3 - 2C_2) = e^x + x$$

Comparing the corresponding co-efficient from both sides.

$$\therefore 2C_1 = 1 \Rightarrow C_1 = 1/2$$

$$C_2 = 1$$

$$C_3 - 2C_2 = 0 \Rightarrow C_3 - 2 = 0 \Rightarrow C_3 = 2$$

$$\therefore y_P = \frac{1}{2}x^2e^x + x + 2$$

Ex. 3.4.3 : Using method of undeterminant co-efficient solve $y'' + 4y = 8x^2$.

**GTU : Summer-23, AEM, Marks 4,
Summer-19,21, Maths-3, Marks 3.5,
Winter-21, AEM, Marks 4,
Winter-23, AEM, Marks 3**

Sol. : Here, $y'' + 4y = 8x^2$ is given

$$\therefore \text{A.E. } D^2 + 4 = 0$$

$$D^2 = -4$$

$$\therefore D = \pm 2i$$

$$\text{C.F.} = C_1 \cos 2x + C_2 \sin 2x$$

Let the trail solution is,

$$y_P = Ax^2 + Bx + C$$

$$\therefore y'_P = 2Ax + B$$

$$y''_P = 2A$$

Put, y''_P and y_P in given equation,

$$\therefore y''_P + 4y_P = 8x^2$$

$$2A + 4Ax^2 + 4Bx + 4C = 8x^2$$

$$\therefore 4Ax^2 + 4Bx + (2A + 4C) = 8x^2$$

Comparing corresponding co-efficient on both sides, we get

$$4A = 8 \Rightarrow A = 2$$

$$4B = 0 \Rightarrow B = 0$$

$$2A + 4C = 0 \Rightarrow C = -1$$

$$\therefore y_P = 2x^2 - 1$$

Now, $y = y_C + y_P = \text{C.F.} + \text{P.I.}$

$$y = C_1 \cos 2x + C_2 \sin 2x + 2x^2 - 1$$

Ex. 3.4.4 : Using method of undetermined multipliers solve $y'' - 3y' + 2y = e^x$.

GTU : Summer-19,21 Maths-3, Marks 3.5

Sol. : Here, $y'' - 3y' + 2y = e^x$ is given

$$\therefore \text{A.E. } D^2 - 3D + 2 = 0$$

$$\therefore D^2 - 2D - D + 2 = 0$$

$$\therefore D(D-2) - 1(D-2) = 0$$

$$(D-1)(D-2) = 0$$

$$\therefore D = 1 \text{ and } D = 2$$

$$\therefore \text{C.F.} = y_C = C_1e^x + C_2e^{2x}$$

Let the trial solution

$$y_P = C_1 e^x$$

Here, e^x term appears in C.F. at once so, we multiply the term corresponding e^x with x .

$$\therefore y_P = C_1 x e^x$$

$$y'_P = C_1 e^x + C_1 x e^x$$

$$y''_P = C_1 e^x + C_1 e^x + C_1 x e^x$$

$$\therefore y''_P = 2C_1 e^x + C_1 x e^x$$

Now, put y_P , y'_P and y''_P in given equation

$$\therefore y''_P - 3y'_P + 2y_P = e^x$$

$$\therefore 2C_1 e^x + C_1 x e^x - 3C_1 e^x - 3C_1 x e^x + 2C_1 x e^x = e^x$$

$$-C_1 e^x = e^x$$

$$\therefore -C_1 = 1 \Rightarrow C_1 = -1$$

$$\therefore y_P = -e^x \cdot x$$

Final solution :

$$\begin{aligned} y &= y_C + y_P \\ &= C_1 e^x + C_2 e^{2x} - e^x \cdot x \end{aligned}$$

Ex. 3.4.5 : Solve $y'' - 5y' + 6y = e^{4x}$.

GTU : Summer-20, Maths-3, Marks 3

Sol. : Here, $y'' - 5y' + 6y = e^{4x}$ is given

$$\therefore \text{A.E. } D^2 - 5D + 6 = 0$$

$$\therefore D^2 - 3D - 2D + 6 = 0$$

$$D(D-3) - 2(D-3) = 0$$

$$\therefore (D-2)(D-3) = 0$$

$$\therefore D = 2 \text{ and } D = 3$$

$$\therefore \text{C.F.} = y_C = C_1 e^{2x} + C_2 e^{3x}$$

Now, here we use U.C. method to find y_P (P.I.)

Let trail solution,

$$y_P = C_1 e^{4x}$$

$$\therefore y'_P = 4C_1 e^{4x}$$

$$\therefore y''_P = 16C_1 e^{4x}$$

Put y_P , y'_P and y''_P in given equation.

$$\therefore y''_P - 5y'_P + 6y_P = e^{4x}$$

$$16C_1 e^{4x} - 20C_1 e^{4x} + 6C_1 e^{4x} = e^{4x}$$

$$2C_1 e^{4x} = e^{4x}$$

$$\therefore 2C_1 = 1 \Rightarrow C_1 = 1/2$$

$$\therefore y_P = \text{P.I.} = \frac{1}{2} e^{4x}$$

Final solution : $y = y_C + y_P$

$$\therefore y = C_1 e^{2x} + C_2 e^{3x} + \frac{1}{2} e^{4x}$$

Ex. 3.4.6 : Using the method of undetermined

co-efficient, solve $y''' + 3y'' + 2y' = x^2 + 4x + 8$.

GTU : Summer-20, Maths-3, Marks 7

Sol. : Here, $y''' + 3y'' + 2y' = x^2 + 4x + 8$ is given

$$\text{A.E. } D^3 + 3D^2 + 2D = 0$$

$$\therefore D(D^2 + 3D + 2) = 0$$

$$\therefore D(D^2 + 2D + D + 2) = 0$$

$$\therefore D(D + (D + 2) + 1(D + 2)) = 0$$

$$\therefore D(D + 2)(D + 1) = 0$$

$$\therefore D = 0, -1, -2$$

$$\therefore \text{C.F.} = y_C = C_1 + C_2 e^{-x} + C_3 e^{-2x}$$

Now, we use U.C. method to find y_P (P.I.)

$\therefore$ Let us consider trail solution,

$$y_P = Ax^2 + Bx + C$$

Here, one of the solution basis 1 is appears in C.F. at once.

So, we multiple trail solution y_P with x .

$$\therefore y_P = Ax^3 + Bx^2 + Cx$$

$$y'_P = 3Ax^2 + 2Bx + C$$

$$y''_P = 6Ax + 2B$$

$$y'''_P = 6A$$

Now, put y_P , y'_P , y''_P and y'''_P in given equation,

$$y'''_P + 3y''_P + 2y'_P = x^2 + 4x + 8$$

$$6A + 18Ax + 6B + 6Ax^2 + 4Bx + 2C = x^2 + 4x + 8$$

$$\therefore 6Ax^2 + (18A + 4B)x + (6A + 6B + 2C) = x^2 + 4x + 8$$

Comparing both side of co-efficient,

$$\therefore 6A = 1 \Rightarrow A = 1/6$$

$$18A + 4B = 4 \Rightarrow 18\left(\frac{1}{6}\right) + 4B = 4$$

$$\Rightarrow 3 + 4B = 4$$

$$\Rightarrow 4B = 1$$

$$\Rightarrow B = 1/4$$

$$6A + 6B + 2C = 8$$

$$6\left(\frac{1}{6}\right) + 6\left(\frac{1}{4}\right) + 2C = 8$$

$$1 + \frac{3}{2} + 2C = 8$$

$$\therefore 2C = 7 - \frac{3}{2}$$

$$\therefore 2C = \frac{11}{2} \Rightarrow C = \frac{11}{4}$$

$$\therefore y_P = \frac{1}{6}x^3 + \frac{1}{4}x^2 + \frac{11}{4}x$$

$$\therefore \text{Final solution : } y = y_C + y_P = C_1 + C_2e^{-x} + C_3e^{-2x} + \frac{1}{6}x^3 + \frac{1}{4}x^2 + \frac{11}{4}x$$

Ex. 3.4.7 : Using the method of undetermined co-efficient solve $y'' - 2y' + 5y = 5x^3 - 6x^2 + 6x$.

GTU : Summer-18, AEM, Marks 4, Summer-23, Maths-3, 7 Marks, Summer-24, Maths-2, Marks 7

Sol. : Here, $y'' - 2y' + 5y = 5x^3 - 6x^2 + 6x$

$$\text{A.E. } D^2 - 2D + 5 = 0$$

Here, $a = 1$, $b = -2$ and $c = 5$

$$\Delta = b^2 - 4ac = 4 - 4(1)(5) = 4 - 20 = -16$$

$$\therefore D_1, D_2 = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 4i}{2}$$

$$D_1, D_2 = 1 \pm 2i$$

$$\therefore \text{C.F.} = y_C = e^x(C_1 \cos 2x + C_2 \sin 2x)$$

Let us consider trial solution,

$$y_P = Ax^3 + Bx^2 + Cx + D$$

$\therefore$

$$y'_P = 3Ax^2 + 2Bx + C$$

$$y''_P = 6Ax + 2B$$

Now, put y_P , y'_P and y''_P in given equation,

$$\therefore y''_P - 2y'_P + 5y_P = 5x^3 - 6x^2 + 6x$$

$$6Ax + 2B - 6Ax^2 - 4Bx - 2C + 5Ax^3 + 5Bx^2 + 5Cx + 5D = 5x^3 - 6x^2 + 6x$$

$$\therefore 5Ax^3 + (5B - 6A)x^2 + (6A - 4B + 5C)x + (2B - 2C + 5D) = 5x^3 - 6x^2 + 6x$$

Comparing co-efficient both side,

$$5A = 5 \Rightarrow A = 1$$

$$5B - 6A = -6 \Rightarrow 5B - 6 = -6$$

$$\Rightarrow 5B = 0 \Rightarrow B = 0$$

$$6A - 4B + 5C = 6 \Rightarrow 6 - 4(0) + 5C = 6$$

$$\Rightarrow 6 + 5C = 6$$

$$\Rightarrow 5C = 0 \Rightarrow C = 0$$

$$2B - 2C + 5D = 0 \Rightarrow 2(0) - 2(0) + 5D = 0$$

$$\Rightarrow D = 0$$

$$\therefore y_P = x^3 - 6x^2$$

$$\therefore y = y_C + y_P$$

$$y = e^x (C_1 \cos 2x + C_2 \sin 2x) + x^3 - 6x^2$$

Ex. 3.4.8 : Solve $(D^2 - 4)y = e^x + \sin 2x$.

GTU : Winter-18, Maths-3, Marks 3

Sol. : Here, $(D^2 - 4)y = e^x + \sin 2x$ is given,

$$\text{A.E.} \quad D^2 - 4 = 0 \Rightarrow D^2 = 4$$

$$\Rightarrow D = \pm 2$$

$$\therefore y_C = C_1 e^{2x} + C_2 e^{-2x}$$

Here, we use U.C. method to find y_P

Let us consider trial solution,

$$y_P = C_1 e^x + C_2 \cos 2x + C_3 \sin 2x$$

$$y'_P = C_1 e^x - 2C_2 \sin 2x + 2C_3 \cos 2x$$

$$y''_P = C_1 e^x - 4C_2 \cos 2x - 4C_3 \sin 2x$$

Put y_p and y_p'' in given equation.

$$\therefore y_p'' - 4y_p = e^x + \sin 2x$$

$$C_1 e^x - 4C_2 \cos 2x - 4C_3 \sin 2x - 4C_1 e^x - 4C_2 \cos 2x - 4C_3 \sin 2x = e^x + \sin 2x$$

$$\therefore -3C_1 e^x - 8C_2 \cos 2x - 8C_3 \sin 2x = e^x + \sin 2x$$

Comparing co-efficient both sides,

$$\therefore -3C_1 = 1 \Rightarrow C_1 = -\frac{1}{3}$$

$$-8C_2 = 0 \Rightarrow C_2 = 0$$

$$-8C_3 = 1 \Rightarrow C_3 = -\frac{1}{8}$$

$$\therefore y_p = -\frac{1}{3}e^x - \frac{1}{8}\sin 2x$$

$$\therefore y = y_C + y_p$$

$$y = C_1 e^{2x} + C_2 e^{-2x} - \frac{1}{3}e^x - \frac{1}{8}\sin 2x$$

Ex. 3.4.9 : Using method of undetermined co-efficient method to solve $y'' + 2y' + 10y = 25x^2 + 3$.

GTU : Winter-19, Maths-3, Marks 3

Sol. : Here, $y'' + 2y' + 10y = 25x^2 + 3$ is given,

$$\text{A.E.} \quad D^2 + 2D + 10 = 0$$

$$\text{Here,} \quad a = 1, b = 2, c = 10$$

$$\therefore \Delta = b^2 - 4ac = 4 - 4(1)(10) = 4 - 40 = -36$$

$$\therefore D_1, D_2 = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-2 \pm \sqrt{-36}}{2} = \frac{-2 \pm 6i}{2} = -1 \pm 3i$$

$$y_C = \text{C.F.} = e^{-x}(C_1 \cos 3x + C_2 \sin 3x)$$

Now, let us consider trial solution

$$y_p = Ax^2 + Bx + C$$

$$y_p' = 2Ax + B$$

$$\therefore y_p'' = 2A$$

Now, put y_p , y_p' and y_p'' in given equation

$$y_p'' + 2y_p' + 10y_p = 25x^2 + 3$$

$$2A + 4Ax + 2B + 10Ax^2 + 10Bx + 10C = 25x^2 + 3$$

Comparing co-efficient both side,

$$10Ax^2 + (4A + 10B)x + (2A + 2B + 10C) = 25x^2 + 3$$

$$\therefore 10A = 25 \Rightarrow A = \frac{25}{10} = \frac{5}{2}$$

$$4A + 10B = 0 \Rightarrow 4\left(\frac{5}{2}\right) + 10B = 0$$

$$\Rightarrow 10 + 10B = 0$$

$$\Rightarrow 10B = -10 \Rightarrow B = -1$$

$$2A + 2B + 10C = 3 \Rightarrow 2\left(\frac{5}{2}\right) + 2(-1) + 10C = 3$$

$$\Rightarrow 5 - 2 + 10C = 3$$

$$\Rightarrow 3 + 10C = 3$$

$$\Rightarrow 10C = 0 \Rightarrow C = 0$$

$$\therefore y_P = \frac{5}{2}x^2 - x$$

$$\therefore y = y_C + y_P$$

$$y = e^{-x}(C_1 \cos 3x + C_2 \sin 3x) + \frac{5}{2}x^2 - x$$

Ex. 3.4.10 : Using method of undetermined co-efficient method to solve $y'' - 9y = x + e^{2x} - \sin 2x$.

GTU : Winter-19, Maths-3, Marks 7

Sol. : Here, $y'' - 9y = x + e^{2x} - \sin 2x$ is given

$$\therefore \text{A.E.} \quad D^2 - 9 = 0 \Rightarrow D^2 = 9$$

$$\Rightarrow D = \mp 3$$

$$\therefore y_C = C_1 e^{3x} + C_2 e^{-3x}$$

Now, let us consider trial solution

$$y_P = Ax + B + C \cdot e^{2x} + D \cos 2x + E \sin 2x$$

$$y'_P = A + 2Ce^{2x} - 2D \sin 2x + 2E \cos 2x$$

$$\therefore y''_P = 4Ce^{2x} - 4D \cos 2x - 4E \sin 2x$$

Now, put y_P and y''_P in given equation

$$\therefore y''_P - 9y_P = x + e^{2x} - \sin 2x$$

$$\therefore 4Ce^{2x} - 4D \cos 2x - 4E \sin 2x - 9Ax - 9B - 9Ce^{2x} - 9D \cos 2x - 9E \sin 2x = x + e^{2x} - \sin 2x$$

Comparing co-efficient both sides,

$$\therefore -5C = 1 \Rightarrow C = -\frac{1}{5}$$

$$-9A = 1 \Rightarrow A = -\frac{1}{9}$$

$$-9B = 0 \Rightarrow B = 0$$

$$-13D = 0 \Rightarrow D = 0$$

$$-13E = -1 \Rightarrow E = \frac{1}{13}$$

$$y_p = -1/9x - 1/5e^{2x} + 1/13 \sin 2x$$

$$y = y_C + y_P$$

$$y = C_1 e^{3x} + C_2 e^{-3x} - \frac{1}{9}x - \frac{1}{5}e^{2x} + \frac{1}{13} \sin 2x$$

Ex. 3.4.11 : Solve the initial value problem $y''' + 3y'' + 3y' + y = 30e^{-x}$, $y(0) = 3$, $y'(0) = -3$, $y''(0) = 47$, by the method of undetermined co-efficients.

GTU : Winter-21, Maths-3, Marks 7

Sol. :

Here , $y''' + 3y'' + 3y' + y = 30e^{-x}$ is given

$$\text{A.E. } (D^3 + 3D^2 + 3D + 1) = 0$$

$$\therefore (D+1)^3 = 0$$

$$\therefore D = -1, -1, -1$$

$$\therefore y_C = e^{-x}(C_1 + C_2x + C_3x^2)$$

Now, let us consider trail solution

$$y_P = C \cdot e^{-x}$$

Here, e^{-x} appears three times in y_C So, we multiply x^3 with trail solution y_P

$$\therefore y_P = Cx^3e^{-x}$$

$$y_P' = 3Cx^2e^{-x} - Cx^3e^{-x}$$

$$y_P'' = 6Cxe^{-x} - 3Cx^2e^{-x} - 3Cx^2e^{-x} + Cx^3e^{-x}$$

$$y_P''' = 6Cxe^{-x} - 6Cx^2e^{-x} + Cx^3e^{-x}$$

$$y_P''' = 6Ce^{-x} - 6Cxe^{-x} - 12Cxe^{-x} + 6Cx^2e^{-x} + 3Cx^2e^{-x} - Cx^3e^{-x}$$

$$\therefore y_P''' = 6Ce^{-x} - 18Cxe^{-x} + 9Cx^2e^{-x} - Cx^3e^{-x}$$

Now, put y_P , y_P' , y_P'' and y_P''' in given equation,

$$\therefore y''' + 3y'' + 3y' + y = 30e^{-x}$$

$$\therefore 6Ce^{-x} - 18Cxe^{-x} + 9Cx^2e^{-x} - Cx^3e^{-x} + 18Cxe^{-x} - 18Cx^2e^{-x} + 3Cx^3e^{-x} + 9Cx^2e^{-x} - 3Cx^3e^{-x} + Cx^3e^{-x} = 30e^{-x}$$

$$\therefore 6Ce^{-x} = 30e^{-x}$$

$$\therefore 6C = 30 \Rightarrow C = 5$$

$$\therefore y_P = 5x^3e^{-x}$$

$$\therefore y = y_C + y_P = e^{-x}(C_1 + C_2x + C_3x^2) + 5x^3e^{-x}$$

$$\therefore y(0) = 3 \Rightarrow y = 3 \text{ when } x = 0$$

∴

$$3 = C_1$$

Now,

$$y_1 = -e^{-x}(C_1 + C_2x + C_3x^2) + e^{-x}(C_2 + 3C_3x) + 15x^2e^{-x} - 5x^3e^{-x}$$

Here,

$$y'(0) = -3 \Rightarrow y' = -3 \text{ when } x = 0$$

$$-3 = -(C_1) + (C_2)$$

$$\Rightarrow -3 = -3 + C_2 \Rightarrow C_2 = 0$$

Now, $y'' =$

$$e^{-x}(C_1 + C_2x + C_3x^2) - e^{-x}(C_2 + 2C_3x) - e^{-x}(C_2 + 3C_3x) + e^{-x}(3C_3) + 30xe^{-x} - 15x^2e^{-x} - 15x^2e^{-x} + 5x^3e^{-x}$$

Here,

$$y''(0) = -47 \Rightarrow y'' = -47 \text{ when } x = 0$$

$$-47 = C_1 - C_2 - C_2 + 3C_3$$

$$-47 = 3 - 0 - 0 + 3C_3$$

$$-49 = 3C_3 \Rightarrow C_3 = -\frac{49}{3}$$

∴

$$y = e^{-x}(3 - 49/3x^2) + 5x^3e^{-x}$$

Ex. 3.4.12 : Solve $(D^2 + 5D + 6)y = e^x$.

GTU : Winter-22, Maths-3, Marks 3

Sol. : Here, $(D^2 + 5D + 6)y = e^x$ is given

A.E.

$$D^2 + 5D + 6 = 0$$

$$D^2 + 3D + 2D + 6 = 0$$

$$D(D + 3) + 2(D + 3) = 0$$

∴

$$(D + 2)(D + 3) = 0$$

∴

$$D = -2, D = -3$$

∴

$$y_C = C_1e^{-2x} + C_2e^{-3x}$$

Let us consider trial solution,

$$y_P = Ce^x$$

$$y'_P = C \cdot e^x$$

$$y''_P = C \cdot e^x$$

Put y_P , y'_P and y''_P in given equation,

$$y''_P + 5y'_P + 6y_P = e^x$$

∴

$$Ce^x + 5Ce^x + 6Ce^x = e^x$$

∴

$$12Ce^x = e^x$$

∴

$$12C = 1 \Rightarrow C = 1/12$$

∴

$$y_P = \frac{1}{12}e^x$$

∴

$$y = y_C + y_P = C_1e^{-2x} + C_2e^{-3x} + \frac{1}{12}e^x$$

Ex. 3.4.13 : Using the method of undetermined co-efficient, find the general solution of the differential equation

$$y'' + 9y = 2x^2$$

GTU : Winter-22, Maths-3, Winter-23, Maths-2, Marks 7

Sol. : Here, $y'' + 9y = 2x^2$

$$\therefore \text{A.E. } D^2 + 9 = 0 \Rightarrow D = \pm 3i$$

$$\therefore y_C = C_1 \cos 3x + C_2 \sin 3x$$

Now, let us consider trail solution y_P

$$\therefore y_P = Ax^2 + Bx + C$$

$$y'_P = 2Ax + B$$

$$y''_P = 2A$$

Put y_P, y'_P and y''_P in given equation,

$$\therefore y''_P + 9y_P = 2x^2$$

$$2A + 9Ax^2 + 9Bx + 9C = 2x^2$$

$$\therefore 9Ax^2 + 9Bx + (2A + 9C) = 2x^2$$

Comparing co-efficient both sides,

$$\therefore 9A = 2 \Rightarrow A = \frac{2}{9}$$

$$9B = 0 \Rightarrow B = 0$$

$$2A + 9C = 0 \Rightarrow 2\left(\frac{2}{9}\right) + 9C = 0$$

$$\Rightarrow \frac{4}{9} + 9C = 0 \Rightarrow 9C = -\frac{4}{9}$$

$$\Rightarrow C = -\frac{4}{81}$$

$$\therefore y_P = \frac{2}{9}x^2 - \frac{4}{81}$$

$$\therefore y = y_C + y_P$$

$$y = C_1 \cos 3x + C_2 \sin 3x + \frac{2}{9}x^2 - \frac{4}{81}$$

Ex. 3.4.14 : Solve $y'' - 3y' + 2y = e^{3x}$

GTU : Summer-18, AEM, Marks 3

Sol. : Here, $y'' - 3y' + 2y = e^{3x}$ is given

$$\therefore \text{A.E. } D^2 - 3D + 2 = 0$$

$$(D - 2)(D - 1) = 0$$

$$\therefore D = 2 \text{ and } D = 1$$

$$y_C = C_1 e^{2x} + C_2 e^x$$

Now, let us consider trail solution,

$$y_P = Ce^{3x}$$

$$y'_P = 3Ce^{3x},$$

$$y''_P = 9Ce^{3x}$$

Put y_P, y'_P and y''_P in given equation.

$$y''_P - 3y'_P + 2y_P = e^{3x}$$

$$9Ce^{3x} - 9Ce^{3x} + 2Ce^{3x} = e^{3x}$$

$$\therefore 2Ce^{3x} = e^{3x} \Rightarrow 2C = 1 \Rightarrow C = 1/2$$

$$\therefore y_P = \frac{1}{2}e^{3x}$$

$$\therefore y = y_C + y_P$$

$$y = C_1 e^{2x} + C_2 e^x + \frac{1}{2}e^{3x}$$

Ex. 3.4.15 : Solve $(D^2 - D - 2)y = \sin 2x$.

GTU : Summer-20, AEM, Marks 4

Sol. : Here, $(D^2 - D - 2)y = \sin 2x$ is given

$$\therefore \text{A.E. } D^2 - D - 2 = 0$$

$$D^2 - 2D + D - 2 = 0$$

$$(D - 2)(D + 1) = 0$$

$$\therefore D = 2, -1$$

$$\therefore y_C = C_1 e^{2x} + C_2 e^{-x}$$

Let us consider trail solution y_P

$$\therefore y_P = A \cos 2x + B \sin 2x$$

$$y'_P = -2A \sin 2x + 2B \cos 2x$$

$$y''_P = -4A \cos 2x - 4B \sin 2x$$

Put, y_P, y'_P and y''_P in given equation

$$\therefore y''_P - y'_P - 2y_P = \sin 2x$$

$$\therefore -4A \cos 2x - 4B \sin 2x + 2A \sin 2x$$

$$- 2B \cos 2x - 2A \cos 2x - 2B \sin 2x = \sin 2x$$

$$\therefore (-6A - 2B) \cos 2x + (2A - 6B) \sin 2x = \sin 2x$$

Comparing co-efficient both sides.

$$\begin{array}{rcl} \therefore -6A - 2B = 0 & \therefore -6A - 2B = 0 \\ 2A - 6B = 1 & \underline{6A - 18B = 1} \\ & -20B = 3 \end{array}$$

$$\therefore -6A + \frac{6}{20} = 0 \quad \Rightarrow \quad B = -\frac{3}{20}$$

$$-6A = -\frac{6}{20}$$

$$\Rightarrow A = \frac{1}{20}$$

$$\therefore y_P = \frac{1}{20} \cos 2x - \frac{3}{20} \sin 2x$$

$$\therefore y = y_C + y_P = C_1 e^{2x} + C_2 e^{-x} + \frac{1}{20} \cos 2x - \frac{3}{20} \sin 2x$$

Ex. 3.4.16 : Using method of undetermined co-efficient, solve $y'' - 2y' = e^x \sin x$.

GTU : Summer-22,24, Marks 7

Sol. : Here, $y'' - 2y' = e^x \sin x$ is given

$$\therefore \text{A.E.} \quad D^2 - 2D = 0$$

$$D(D - 2) = 0$$

$$\therefore D = 0 \text{ and } D = 2$$

$$\therefore y_C = \text{C.F.} = C_1 e^{0x} + C_2 e^{2x}$$

$$\text{C.F.} = C_1 + C_2 e^{2x}$$

Let trial solution of y'_P

$$y_P = e^x (A \cos x + B \sin x)$$

$$y'_P = e^x (-A \sin x + B \cos x) + e^x (A \cos x + B \sin x)$$

$$y''_P = e^x (-A \sin x + B \cos x) + e^x (-A \cos x - B \sin x) + e^x (A \cos x + B \sin x) + e^x (-A \sin x + B \cos x)$$

$$\therefore y''_P = 2e^x (-A \sin x + B \cos x)$$

$$y''_P = -2A e^x \sin x + 2B e^x \cos x$$

Now, put y'_P and y''_P in given equation.

$$y''_P - 2y'_P = e^x \sin x$$

$$-2A e^x \sin x + 2B e^x \cos x + 2A e^x \sin x - 2B e^x \cos x - 2A e^x \cos x - 2B e^x \sin x = e^x \sin x$$

$$-2A e^x \cos x - 2B e^x \sin x = e^x \sin x$$

$$-2A = 0, \quad -2B = 1$$

$$A = 0 \text{ and } B = -\frac{1}{2}$$

$$\therefore y_P = e^x \left(0 \cdot \cos x - \frac{1}{2} \sin x \right) = -\frac{1}{2} e^x \sin x$$

$$\therefore y_P = y_C + y_P = C_1 + C_2 e^{2x} - \frac{1}{2} e^x \sin x$$

Ex. 3.4.17 : Solve $y'' + y' = 5e^x - \sin 2x$.

GTU : Summer-19, AEM, Marks 4

Sol. : Here, $y'' + y' = 5e^x - \sin 2x$ is given,

$$\text{A.E.} \quad D^2 + D = 0$$

$$\therefore D(D+1) = 0$$

$$\therefore D = 0 \text{ and } D = -1$$

$$\therefore y_C = C_1 e^{0x} + C_2 e^{-x}$$

$$\therefore y_C = C_1 + C_2 e^{-x}$$

Let us consider trial solution y_P

$$\therefore y_P = C \cdot e^x + A \cos 2x + B \sin 2x$$

$$y'_P = C \cdot e^x - 2A \sin 2x + 2B \cos 2x$$

$$y''_P = C \cdot e^x - 4A \cos 2x + 4B \sin 2x$$

Put, y_P , y'_P and y''_P in given equation.

$$y''_P + y'_P = 5e^x - \sin 2x$$

$$C \cdot e^x - 4A \cos 2x - 4B \sin 2x + C \cdot e^x - 2A \sin 2x + 2B \cos 2x = 5e^x - \sin 2x$$

$$\therefore 2C \cdot e^x + (2B - 4A) \cos 2x + (-2A - 4B) \sin 2x = 5e^x - \sin 2x$$

Comparing co-efficient both sides,

$$2C = 5 \Rightarrow C = 5/2$$

$$2B - 4A = 0 \quad \therefore -4A + 2B = 0$$

$$-2A - 4B = -1 \quad -4A - 8B = -2$$

$$+ \quad + \quad +$$

$$10B = 2$$

$$\therefore 2 \left(\frac{1}{5} \right) - 4A = 0 \quad \therefore B = \frac{1}{5}$$

$$\therefore 4A = \frac{2}{5} \Rightarrow A = \frac{1}{10}$$

$$\therefore y_P = \frac{5}{2} e^x + \frac{1}{10} \cos 2x + \frac{1}{5} \sin 2x$$

$$\therefore y = y_C + y_P$$

Ex. 3.4.18 : Solve $y'' - 2y' + y = 10 \cdot e^x$

**GTU : Summer-22, AEM, Marks 4,
Winter-22, Maths-2, Marks 4**

Sol. : Here, $y'' - 2y' + y = 10 \cdot e^x$ is given

$$\text{A.E. } D^2 - 2D + 1 = 0$$

$$\therefore (D-1)^2 = 0$$

$$\therefore D = 1, 1$$

$$\therefore y_C = e^x(C_1 + C_2 x)$$

Now, consider trial solution of y_P

$$y_P = C \cdot e^x$$

Here, e^x term appears twice in y_C

$\therefore$ We multiply x^2 in corresponding terms,

$$\therefore y_P = C \cdot x^2 e^x$$

$$y'_P = 2Cex + Cx^2 e^x$$

$$y''_P = 2Cex + 2Cxe^x + 2Cxe^x + Cx^2 e^x$$

$$\therefore y''_P = 2Cex + 4Cxe^x + Cx^2 e^x$$

Now, put y_P , y'_P and y''_P in given equation,

$$y''_P - 2y'_P + y_P = 10e^x$$

$$\therefore 2Cex + 4Cxe^x + Cx^2 e^x - 4Cxe^x - 2Cx^2 e^x + Cx^2 e^x = 10e^x$$

$$\therefore 2Cex = 10e^x \therefore 2C = 10 \Rightarrow C = 5$$

$$\therefore y_P = 5x^2 e^x$$

$$\therefore y = y_C + y_P = e^x(C_1 + C_2 x) + 5x^2 e^x$$

Ex. 3.4.19 : Find the solution of $y'' + 4y = 2 \sin 3x$ by the method of undetermined co-efficients.

**GTU : Summer-23, AEM, Marks 7,
Winter-22, Maths-2, Marks 7**

Sol. : Here, $y'' + 4y = 2 \sin 3x$ is given,

$$\text{A.E. } D^2 + 4 = 0$$

$$\therefore D^2 = -4$$

$$\therefore D = \pm 2i$$

$$\therefore y_C = C_1 \cos 2x + C_2 \sin 2x$$

Let us consider trial solution y_P

$$\therefore y_P = A \cos 3x + B \sin 3x$$

$$y'_P = -3A \sin 3x + 3B \cos 3x$$

$$y''_P = -9A \cos 3x - 9B \sin 3x$$

Put, y_P and y''_P in given equation,

$$y''_P + 4y_P = 2 \sin 3x$$

$$\therefore -9A \cos 3x - 9B \sin 3x + 4A \cos 3x + 4B \sin 3x = 2 \sin 3x$$

$$\therefore -5A \cos 3x - 5B \sin 3x = 2 \sin 3x$$

Comparing co-efficient both sides,

$$\therefore -5A = 0 \Rightarrow A = 0$$

$$-5B = 2 \Rightarrow B = -\frac{2}{5}$$

$$\therefore y_P = -\frac{2}{5} \sin 2x$$

$$\therefore y = y_C + y_P$$

$$= C_1 \cos 2x + C_2 \sin 2x - \frac{2}{5} \sin 2x$$

Ex. 3.4.20 : Using the method of undetermined co-efficient find the general solution of $y'' - 4y' - 12y = 8x^2$.

GTU : Summer-24, AEM, Marks 7

Sol. : Here, $y'' - 4y' - 12y = 8x^2$ is given.

$$\therefore \text{A.E. } D^2 - 4D - 12 = 0$$

$$D^2 - 6D + 2D - 12 = 0$$

$$\therefore D(D - 6) + 2(D - 6) = 0$$

$$\therefore (D + 2)(D - 6) = 0$$

$$\therefore D = -2, D = 6$$

$$\therefore y_C = C_1 e^{-2x} + C_2 e^{6x}$$

Now, let us consider trial solution y_P

$$\therefore y_P = Ax^2 + Bx + C$$

$$y'_P = 2Ax + B$$

$$y''_P = 2A$$

Put, y_P , y'_P and y''_P in given equation.

$$\begin{aligned} \therefore y_P'' - 4y_P' - 12y_P &= 8x^2 \\ \therefore 2A - 8Ax - 4B - 12Ax^2 - 12Bx - 12C &= 8x^2 \\ \therefore -12Ax^2 + (-8A - 12B)x + (2A - 4B - 12C) &= 8x^2 \end{aligned}$$

Comparing co-efficient both sides.

$$\therefore -12A = 8 \Rightarrow A = -\frac{8}{12} \Rightarrow A = -\frac{2}{3}$$

$$\therefore -8A - 12B = 0 \Rightarrow (-8)\left(-\frac{2}{3}\right) - 12B = 0$$

$$\Rightarrow 12B = \frac{16}{3} \Rightarrow B = \frac{16}{3 \times 12} = \frac{4}{9}$$

$$\therefore 2A - 4B - 12C = 0$$

$$\therefore 2\left(-\frac{2}{3}\right) - 4\left(\frac{4}{9}\right) - 12C = 0$$

$$-\frac{4}{3} - \frac{16}{9} - 12C = 0$$

$$\therefore -\frac{84}{27} = 12C$$

$$\therefore C = -\frac{7}{27}$$

$$\therefore y_P = -\frac{2}{3}x^2 + \frac{4}{9}x - \frac{7}{27}$$

$$\therefore y = y_C + y_P$$

$$\therefore y = C_1 e^{-2x} + C_2 e^{6x} - \frac{2}{3}x^2 + \frac{4}{9}x - \frac{7}{27}$$

Ex. 3.4.21 : Solve $(D^2 - 4)y = 1 + e^x$.

GTU : Winter-18, AEM, Marks 4

Sol. : $(D^2 - 4)y = 1 + e^x$ is given

$$\therefore \text{A.E. } D^2 - 4 = 0$$

$$D^2 = 4$$

$$\therefore D = \pm 2$$

$$\therefore y_C = C_1 e^{2x} + C_2 e^{-2x}$$

For y_P , we use U.C. method

Let us take trial solution y_P

$$\therefore y_P = C + C_1 e^x$$

$$y_P' = C_1 e^x \text{ and } y_P'' = C_1 e^x$$

Put y_P , y_P' and y_P'' in given equation,

$$y_P'' - 4y_P' = 1 + e^x$$

$$C_1 e^x - 4C - 4C_1 e^x = 1 + e^x$$

$$-4C - 3C_1 e^x = 1 + e^x$$

Comparing co-efficient both side,

$$-4C = 1 \Rightarrow C = -\frac{1}{4}$$

$$-3C_1 = 1 \Rightarrow C_1 = -\frac{1}{3}$$

$$\therefore y_P = -\frac{1}{4} - \frac{1}{3}e^x$$

$$y = y_C + y_P$$

$$\therefore y = C_1 e^{2x} + C_2 e^{-2x} - \frac{1}{4} - \frac{1}{3}e^x$$

Ex. 3.4.22 : Solve by the method of undetermined co-efficient $y'' + 10y' + 25y = e^{-5x}$.

GTU : Winter-18, AEM, Marks 4

Sol. : Here, $y'' + 10y' + 25y = e^{-5x}$ is given,

$$\therefore \text{A.E. } D^2 + 10D + 25 = 0$$

$$\therefore (D+5)^2 = 0$$

$$\therefore D = -5, -5$$

$$\text{C.F.} = y_C$$

$$= e^{-5x}(C_1 + C_2 x)$$

Let us consider trial solution y_P

$$\therefore y_P = C \cdot e^{-5x}$$

Here, e^{-5x} appears twice in y_C

$\therefore$ We multiply x^2 in corresponding term y_P .

$$\therefore y_P = C x^2 e^{-5x}$$

$$\therefore y_P = 2C x e^{-5x} - 5C x^2 e^{-5x}$$

$$y_P'' = 2C e^{-5x} - 10C x e^{-5x} - 10C x e^{-5x} + 25C x^2 e^{-5x}$$

$$\therefore y_P'' = 2C e^{-5x} - 20C x e^{-5x} + 25C x^2 e^{-5x}$$

Now, put y_P , y_P' and y_P'' in given equation.

$$y'' + 10y' + 25y = e^{-5x}$$

$$2C e^{-5x} - 20C x e^{-5x} + 25C x^2 e^{-5x} + 20C x e^{-5x} - 50C x^2 e^{-5x} + 25C x^2 e^{-5x} = e^{-5x}$$

$$\therefore 2C e^{-5x} = e^{-5x}$$

$$\therefore 2C = 1 \Rightarrow C = 1/2$$

$$\therefore y_P = \frac{1}{2} x^2 e^{-5x}$$

$$\therefore y = y_C + y_P = e^{-5x} (C_1 + C_2 x) + \frac{1}{2} x^2 e^{-5x}$$

Ex. 3.4.23 : Using undetermined co-efficient method solve the differential equation $y'' + y' - 6y = 6x + 3x^2 - 6x^3$.

GTU : Winter-19, AEM, Marks 7

Sol. : Here, $y'' + y' - 6y = 6x + 3x^2 - 6x^3$ is given

$$\therefore \text{A.E.} \quad D^2 + D - 6 = 0$$

$$\therefore D^2 + 3D - 2D - 6 = 0$$

$$D(D + 3) - 2(D + 3) = 0$$

$$(D - 2)(D + 3) = 0$$

$$D = 2 \text{ and } D = -3$$

$$\therefore \text{C.F.} = y_C = C_1 e^{2x} + C_2 e^{-3x}$$

Let us consider trail solution y_P

$$y_P = Ax^3 + Bx^2 + Cx + D$$

$$y'_P = 3Ax^2 + 2Bx + C$$

$$y''_P = 6Ax + 2B$$

Put y_P , y'_P and y''_P in given equation,

$$\therefore y''_P + y'_P - 6y_P = 6x + 3x^2 - 6x^3$$

$$(6Ax + 2B) + (3Ax^2 + 2Bx + C) - 6(Ax^3 + Bx^2 + Cx + D) = 6x + 3x^2 - 6x^3$$

$$\therefore -6Ax^3 + (3A - 6B)x^2 + (6A + 2B - 6C)x + (2B + C - 6D) = 6x + 3x^2 - 6x^3$$

Comparing co-efficient both sides,

$$\therefore -6A = -6 \Rightarrow A = 1$$

$$3A - 6B = 3 \Rightarrow 4 - 6B = 3 \Rightarrow B = 0$$

$$6A + 2B - 6C = 6$$

$$\Rightarrow 6(1) + 2(0) - 6C = 6$$

$$\Rightarrow C = 0$$

$$2B + C - 6D = 0$$

$$2(0) + (0) - 6D = 0$$

$$\Rightarrow D = 0$$

$$y_P = x^3$$

$$y = y_C + y_P = C_1 e^{2x} + C_2 e^{-3x} + x^3$$

Ex. 3.4.24 : Solve $y'' + 5y' + 6y = 12e^x$.

GTU : Winter-21, AEM, Marks 4

Sol. : Here, $y'' + 5y' + 6y = 12e^x$ is given

$$\therefore \text{A.E. } D^2 + 5D + 6 = 0$$

$$(D + 3)(D + 2) = 0$$

$$D = -3, D = -2$$

$$\text{C.F.} = y_C = C_1 e^{-3x} + C_2 e^{-2x}$$

Now, for y_P we use U.C. method

Let us consider trail solution y_P

$$y_P = Ce^x$$

$$y'_P = C \cdot e^x$$

$$y''_P = C \cdot e^x$$

Put y_P, y'_P and y''_P in given equation,

$$\therefore y''_P + 5y'_P + 6y_P = 12e^x$$

$$Ce^x + 5Ce^x + 6Ce^x = 12e^x$$

$$12Ce^x = 12e^x$$

$$\therefore 12C = 12 \Rightarrow C = 1$$

$$\therefore y_P = e^x$$

$$y = y_C + y_P = C_1 e^{-3x} + C_2 e^{-2x} + e^x$$

Ex. 3.4.25 : Solve $y'' + 6y' + 9y = e^{3x}$.

GTU : Winter-21, AEM, Marks 3

Sol. : Here, $y'' + 6y' + 9y = e^{3x}$ is given,

$$\therefore \text{A.E. } D^2 + 6D + 9 = 0$$

$$(D + 3)^2 = 0$$

$$\therefore D = -3, -3$$

$$\text{C.F.} = y_C = e^{-3x}(C_1 + C_2 x)$$

Let us consider trail solution y_P

$$\therefore y_P = C \cdot e^{3x}$$

$$y'_P = 3Ce^{3x}$$

$$y''_P = 9Ce^{3x}$$

Put y_P, y'_P and y''_P in given equation.

$$\therefore y''_P + 6y'_P + 9y_P = e^{3x}$$

$$9Ce^{3x} + 18Ce^{3x} + 9Ce^{3x} = e^{3x}$$

$$36Ce^{3x} = e^{3x}$$

$$\therefore 36C = 1 \Rightarrow C = \frac{1}{36}$$

$$\therefore y_P = \frac{1}{36}e^{3x}$$

$$\therefore y = y_C + y_P = e^{-3x}(C_1 + C_2 x) + \frac{1}{36}e^{3x}$$

Ex. 3.4.26 : Solve $(D^3 - D^2 - 6D)y = x^2 + 1$.

GTU : Winter-22, AEM, Marks 4

Sol. : Here, $(D^3 - D^2 - 6D)y = x^2 + 1$ is given

$$\therefore \text{A.E. } (D^3 + D^2 + 6D)y = 0$$

$$D(D^2 - D - 6) = 0$$

$$\therefore D(D^2 - 3D + 2D - 6) = 0$$

$$D(D - 3)(D + 2) = 0$$

$$D = 0, 3, -2$$

$$\therefore \text{C.F.} = y_C = C_1 + C_2 e^{3x} + C_3 e^{-2x}$$

Here, we use U.C. Method to find y_P

$\therefore$ Let us consider trail solution y_P

$$y_P = Ax^2 + Bx + C$$

Here term constant appears once in y_C

$\therefore$ Multiply x with y_P

$$\therefore y_P = Ax^3 + Bx^2 + Cx$$

$$y'_P = 3Ax^2 + 2Bx + C$$

$$y''_P = 6Ax + 2B$$

$$y'''_P = 6A$$

put y'_P, y''_P and y'''_P in given equation.

$$\therefore y'''_P - y''_P - 6y'_P = x^2 + 1$$

$$6A - 6Ax - 2B - 18Ax^2 - 12Bx - 6C = x^2 + 1$$

$$-18Ax^2 + (-6A - 12B)x + (6A - 2B - 6C) = x^2 + 1$$

Comparing co-efficient both sides.

$$-18A = 1 \Rightarrow A = -\frac{1}{18}$$

$$-6A - 12B = 0 \Rightarrow B = 0$$

$$6A - 2B - 6C = 1$$

$$6\left(-\frac{1}{18}\right) - 2(0) - 6C = 1$$

$$-\frac{1}{3} - 6C = 1$$

$$\Rightarrow 6C = -\frac{1}{3} - 1 = -\frac{4}{3}$$

$$\Rightarrow C = -\frac{4}{18}$$

$$\therefore y_P = -\frac{1}{18}x^3 - \frac{4}{18}$$

$$\therefore y = y_C + y_P$$

$$y = C_1 + C_2 e^{3x} + C_3 e^{-2x} - \frac{1}{18}x^3 - \frac{4}{18}$$

Ex. 3.4.27 : Use the method of undetermined co-efficient to solve differential equation $y'' - 2y' + y = x^2 e^x$.

GTU : Summer-19, 20, Maths-2, Marks 7, Summer-23, Maths-2, Marks 3.5

Sol. : Here, $y'' - 2y' + y = x^2 e^x$ is given

A.E.

$$D^2 - 2D + 1 = 0$$

$$(D-1)^2 = 0$$

$$D = 1, 1$$

$\therefore$ C.F.

$$y_C = (C_1 + C_2 x)e^x$$

Let us consider trial solution y_P

$$\therefore y_P = e^x(Ax^2 + Bx + C)$$

Now, e^x term appears twice in y_C multiply x^2 in y_P

$$\therefore y_P = e^x(Ax^4 + Bx^3 + Cx^2)$$

$$\therefore y'_P = e^x(Ax^4 + Bx^3 + Cx^2) + e^x(4Ax^3 + 3Bx^2 + 2Cx)$$

$$y''_P = e^x(Ax^4 + Bx^3 + Cx^2) + e^x(4Ax^3 + 3Bx^2 + 2Cx) + e^x(4Ax^3 + 3Bx^2 + 2Cx) + e^x(12Ax^2 + 6Bx + 2C)$$

$$\therefore y''_P = e^x(Ax^4 + Bx^3 + Cx^2) + 2e^x(4Ax^3 + 3Bx^2 + 2Cx) + e^x(12Ax^2 + 6Bx + 2C)$$

Put y_P , y'_P and y''_P in given equation,

$$y''_P - 2y'_P + y_P = x^2 e^x$$

$$e^x(Ax^4 + Bx^3 + Cx^2) + 2e^x(4Ax^3 + 3Bx^2 + 2Cx) + e^x(12Ax^2 + 6Bx + 2C)$$

$$-2e^x(Ax^4 + Bx^3 + Cx^2) - 2e^x(4Ax^3 + 3Bx^2 + 2Cx) + e^x(Ax^4 + Bx^3 + Cx^2) = x^2 e^x$$

$$e^x(12Ax^2 + 6Bx + 2C) = x^2 e^x$$

Now compare co-efficient both side,

$$12A = 1, 6B = 0, 2C = 0$$

$$A = \frac{1}{12}, B = 0, C = 0$$

$$y_P = \frac{1}{12} e^x x^4$$

$$y = y_C + y_P$$

$$y = (C_1 + C_2 x) e^x + \frac{1}{12} e^x x^4$$

Ex. 3.4.28 : Solve $y'' - 2y' + y = \cos 3x$.

GTU : Summer-22, Maths-2, Marks 3

Sol. : Here, $y'' - 2y' + y = \cos 3x$ is given

A.E.

$$D^2 - 2D + 1 = 0$$

$$(D-1)^2 = 0$$

$$D = 1, 1$$

$$C.F. = y_C e^x (C_1 + C_2 x)$$

Now, let trail solution y_P

$$y_P = A \cos 3x + B \sin 3x$$

$$y'_P = -3A \sin 3x + 3B \cos 3x$$

$$y''_P = -9A \cos 3x - 9B \sin 3x$$

Put, y_P , y'_P and y''_P in given equation

$$y''_P - 2y'_P + y = \cos 3x$$

$$-9A \cos 3x - 9B \sin 3x + 6A \sin 3x - 6B \cos 3x + A \cos 3x + B \sin 3x = \cos 3x$$

$$-8A \cos 3x - 8B \sin 3x + 6A \sin 3x - 6B \cos 3x = \cos 3x$$

$$\therefore (-8A - 6B) \cos 3x + (6A - 8B) \sin 3x = \cos 3x$$

$$\therefore (-8A - 6B) \cos 3x + (6A - 8B) \sin 3x = \cos 3x$$

Comparing co-efficient both sides,

$$-8A - 6B = 1$$

$$6A - 8B = 0$$

$$\therefore -48A - 36B = 6$$

$$48A - 64B = 0$$

$$-100B = 6 \Rightarrow B = -\frac{6}{100}$$

$$6A - 8\left(-\frac{3}{50}\right) = 0 \Rightarrow B = -\frac{3}{50}$$

$$6A + \frac{24}{50} = 0$$

$$\therefore 6A = -\frac{24}{50} \Rightarrow A = -\frac{24}{6 \times 50}$$

$$\Rightarrow A = -\frac{4}{50}$$

$$\therefore y_P = -\frac{4}{50} \cos 3x - \frac{3}{50} \sin 3x$$

$$\therefore y_P = y_C + y_P$$

$$y = e^x (C_1 + C_2 x) - \frac{4}{50} \cos 3x - \frac{3}{50} \sin 3x$$

Ex. 3.4.29 : Using method of undetermined co-efficient obtain the solution of $y'' + 4y = \sin x$.

GTU : Summer-23, Maths-2, Marks 4

Sol. : Here, $y'' + 4y = \sin x$

A.E. $D^2 + 4 = 0 \Rightarrow D^2 = -4 \Rightarrow D = \pm 2i$

$$y_C = C_1 \cos 2x + C_2 \sin 2x$$

Now, let us consider trial solution y_P

$$y_P = A \cos x + B \sin x$$

$$y'_P = -A \sin x + B \cos x$$

$$y''_P = -A \cos x - B \sin x$$

Put y_P , y'_P and y''_P in given equation

$$y''_P + 4y_P = \sin x$$

$$-A \cos x - B \sin x + 4A \cos x + 4B \sin x = \sin x$$

$$3A \cos x + 3B \sin x = \sin x$$

$$\therefore 3A = 0 \Rightarrow A = 0$$

$$3B = 1 \Rightarrow B = \frac{1}{3}$$

$$\therefore y_P = \frac{1}{3} \sin x,$$

$$y = y_C + y_P = C_1 \cos 2x + C_2 \sin 2x + \frac{1}{3} \sin x$$

Ex. 3.4.30 : Solve $(D^2 + 9)y = 2 \sin 3x + \cos 3x$.

GTU : Summer-24, Maths 2, Marks 4

Sol. : Here, $(D^2 + 9)y = 2 \sin 3x + \cos 3x$

A.E. $D^2 + 9 = 0$

$$\therefore D^2 = -9 \Rightarrow D = \pm 3i$$

$$\therefore y_C = C_1 \cos 3x + C_2 \sin 3x$$

Let $y_P = C_1 \cos 3x + C_2 \sin 3x$

Here, $\cos 3x$ and $\sin 3x$ appears in y_C

$$y_P = x(C_1 \cos 3x + C_2 \sin 3x)$$

$$y'_P = (C_1 \cos 3x + C_2 \sin 3x) + x(-3C_1 \sin 3x + 3C_2 \cos 3x)$$

$$y''_P = (-3C_1 \sin 3x + 3C_2 \cos 3x) + (-3C_1 \sin 3x + 3C_2 \cos 3x) + x(-9C_1 \cos 3x - 9C_2 \sin 3x)$$

Put y_P , y'_P and y''_P in given equation

$$y''_P + 9y_P = 2 \sin 3x$$

Ex. 3.4.31 : Solve $(D^2 + 3D + 2)y = \sin 2x$.

Sol. : Here, $(D^2 + 3D + 2)y = \sin 2x$ is given

A.E.

$$D^2 + 3D + 2 = 0$$

$$D^2 + 2D + D + 2 = 0$$

$$D(D + 2) + 1(D + 2) = 0$$

$$(D + 1)(D + 2) = 0$$

$$D = -1, -2$$

$$y_C = C_1 e^{-x} + C_2 e^{-2x}$$

Let us consider trial solution y_P

$$y_P = A \cos 2x + B \sin 2x$$

$$y'_P = -2A \sin 2x + 2B \cos 2x$$

$$y''_P = -4A \cos 2x - 4B \sin 2x$$

Put y_P , y'_P and y''_P in given equation

$$(y''_P + 3y'_P + 2y_P) = \sin 2x$$

$$-4A \cos 2x - 4B \sin 2x - 6A \sin 2x + 6B \cos 2x + 2A \cos 2x + 2B \sin 2x = \sin 2x$$

$$-2A \cos 2x - 2B \sin 2x - 6A \sin 2x + 6B \cos 2x = \sin 2x$$

$$\therefore (-2A + 6B) \cos 2x + (-2B - 6A) \sin 2x = \sin 2x$$

Comparing co-efficient

$$-2A + 6B = 0 \quad -6A + 18B = 0$$

$$-6A - 2B = 1 \quad -6A - 2B = 1$$

$$\begin{array}{r} + \quad + \quad - \\ \hline \end{array}$$

$$-2A + 6\left(\frac{-1}{20}\right) = 0 \quad 20B = -1$$

$$B = \frac{-1}{20}$$

$$\therefore -2A = \frac{6}{20} \Rightarrow A = \frac{-3}{20}$$

$$y_P = \frac{-3}{20} \cos 2x - \frac{1}{20} \sin 2x$$

$$y = y_C + y_P$$

$$y = C_1 e^{-x} + C_2 e^{-2x} - \frac{3}{20} \cos 2x - \frac{1}{20} \sin 2x$$

Ex. 3.4.32 : Using undetermined co-efficient method solve $(D^2 + 4D)y = x + x^2$.

GTU : Winter-23, Maths-2, Marks 3

Sol. : Here, given $(D^2 + 4D)y = x + x^2$

A.E. $D^2 + 4D = 0$

$$D(D + 4) = 0$$

$$D = 0, D = -4$$

$$y_C = \text{C.F.} = C_1 + C_2 e^{-4x}$$

Let us consider trial solution y_P .

$$y_P = Ax^2 + Bx + C$$

Here, constant term appear in y_C .

$$y_P = Ax^3 + Bx^2 + Cx$$

$$y'_P = 3Ax^2 + 2Bx + C$$

$$y''_P = 6Ax + 2B$$

$\therefore y_P, y'_P$ and y''_P in given equation,

$$y''_P + 4y'_P = x + x^2$$

$$6Ax + 2B + 12Ax^2 + 8Bx + 4C = x + x^2$$

$$\therefore 12Ax^2 + (6A + 8B)x + (2B + 4C) = x + x^2$$

$$\therefore 12A = 1 \Rightarrow A = \frac{1}{12}$$

$$6A + 8B = 1 \Rightarrow 6\left(\frac{1}{12}\right) + 8B = 1 \Rightarrow \frac{1}{2} + 8B = 1$$

$$\therefore 8B = 1 - \frac{1}{2} \Rightarrow 8B = \frac{1}{2} \Rightarrow B = \frac{1}{16}$$

$$2B + 4C = 0 \Rightarrow 2\left(\frac{1}{16}\right) + 4C = 0 \Rightarrow \frac{1}{8} + 4C = 0$$

$$\Rightarrow 4C = -\frac{1}{8} \Rightarrow C = -\frac{1}{32}$$

$$\therefore y_P = \frac{1}{12}x^3 + \frac{1}{16}x^2 - \frac{1}{32}x$$

$$\therefore y = y_C + y_P$$

$$y = C_1 + C_2 e^{-4x} + \frac{1}{12}x^3 + \frac{1}{16}x^2 - \frac{1}{32}x$$

Ex. 3.4.33 : Solve $y'' + y' - 12y = e^{6x}$.

Sol. : Here, $y'' + y' - 12y = e^{6x}$ is given

A.E.

$$(D^2 + D - 12) = 0$$

$$D^2 + 4D - 3D - 12 = 0$$

$$D(D + 4) - 3(D + 4) = 0$$

$$D = 3, D = -4$$

$$y_C = C_1 e^{3x} + C_2 e^{-4x}$$

Let us consider trial solution y_P

$$\therefore y_P = C \cdot e^{6x}, y'_P = 6C e^{6x}$$

$$y''_P = 36C e^{6x}$$

Put y_P, y'_P and y''_P in given equation

$$\therefore y''_P + y'_P - 12y_P = e^{6x}$$

$$36C e^{6x} + 6C e^{6x} - 12C e^{6x} = e^{6x}$$

$$30C e^{6x} = e^{6x}$$

$$\therefore 30C = 1 \Rightarrow C = \frac{1}{30}$$

$$\therefore y_P = \frac{1}{30} e^{6x}$$

$$\therefore y = C_1 e^{3x} + C_2 e^{-4x} + \frac{1}{30} e^{6x}$$

Ex. 3.4.34 : Solve $y'' - 4y' - 12y = \sin x$ by method of undetermined co-efficient.

GTU : Summer-20, AEM, Marks 4

Sol. : Here, $y'' - 4y' - 12y = \sin x$ is given,

$$\text{A.E. } D^2 - 4D - 12 = 0$$

$$D^2 - 6D + 2D - 12 = 0$$

$$\therefore D(D - 6) + 2(D - 6) = 0$$

$$(D + 2)(D - 6) = 0$$

$$\therefore D = -2 \text{ and } D = 6$$

$$\therefore y_C = C_1 e^{-2x} + C_2 e^{6x}$$

Now, let us consider trial solution.

$$y_P = A \cos x + B \sin x$$

$$y'_P = -A \sin x + B \cos x$$

$$y''_P = -A \cos x - B \sin x$$

Put y_P , y'_P and y''_P in given equation.

$$y''_P - 4y'_P - 12y_P = \sin x$$

$$-A \cos x - B \sin x + 4A \sin x - 4B \cos x - 12A \cos x - 12B \sin x = \sin x$$

$$-13A \cos x - 4B \cos x - 13B \sin x + 4A \sin x = \sin x$$

$$(-13A - 4B) \cos x + (4A - 13B) \sin x = \sin x$$

Comparing co-efficient both sides,

$$\begin{cases} -13A - 4B = 0 \\ 4A - 13B = 1 \end{cases} \Rightarrow \begin{cases} -52A - 16B = 0 \\ 52A - 169B = 13 \end{cases}$$

$$\underline{-185B = 13}$$

$$\therefore B = -\frac{13}{185}$$

$$\therefore 4A - 13\left(-\frac{13}{185}\right) = 1$$

$$\Rightarrow 4A = 1 - \frac{169}{185} \Rightarrow 4A = \frac{16}{185}$$

$$\Rightarrow A = \frac{4}{185}$$

$$\therefore y_P = \frac{4}{185} \cos x - \frac{13}{185} \sin x$$

$$\therefore y = y_C + y_P = C_1 e^{-2x} + C_2 e^{6x} + \frac{4}{185} \cos x - \frac{13}{185} \sin x$$

3.5 Method of Variation of Parameters

GTU : Summer-12,18,19,20,21,22,23,24,
Winter-18,19,21,22,23, Marks 7

- **Method of variation of parameters** : Let us consider a differential equation of the form $f(D) y = r(x)$.
- When $r(x)$ is the form e^{ax} , $\sin ax$, $\cos ax$, x^m then we use undetermined co-efficient method for finding P.I. (y_P). If $r(x)$ be of any other form like $\tan x$, $\sec x$, $\operatorname{cosec} x$, $\cot x$ etc., then we have to use method of variation of parameters for finding y_P .

Consider second order linear ODE with constant co-efficient,

$$y'' + C_1 y' + C_2 y = r(x) \quad \dots(3.5.1)$$

The C.F. (Complementary Function) of equation (3.5.1) is

$$\text{C.F.} = y_C = C_1 y_1 + C_2 y_2$$

Where, y_1 and y_2 will satisfy the equation

$$y'' + C_1 y' + C_2 y = 0$$

Now for finding P.I. (y_P) first find Wronskin's W.

If $W \neq 0$ then y_1 and y_2 are linearly independent.

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$W_1 = \begin{vmatrix} 0 & y_2 \\ 1 & y_2' \end{vmatrix} = -y_2$$

$$W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & 1 \end{vmatrix} = y_1$$

$$y_P = y_1 \int \frac{W_1}{W} r(x) dx + y_2 \int \frac{W_2}{W} r(x) dx$$

If differential equation of the form

$$\frac{d^3 y}{dx^3} + C_1 \frac{d^2 y}{dx^2} + C_2 \frac{dy}{dx} + C_3 = r(x) \quad \dots(3.5.2)$$

The C.F. (Complimentary Function) of equation (3.5.2) is

$$C.F. = y_C = C_1 y_1 + C_2 y_2 + C_3 y_3$$

$$\therefore \text{Wronskin's } W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix}$$

and

$$W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix}$$

$$\therefore y_P = y_1 \int \frac{W_1}{W} r(x) dx + y_2 \int \frac{W_2}{W} r(x) dx + y_3 \int \frac{W_3}{W} r(x) dx$$

Ex. 3.5.1 : Using the method of variation of parameter solve the differential equation : $(D^2 + 2D + 1)y = e^{-x} \cdot \log x$.

GTU : Summer-18, Maths-3, Marks 7

Sol. : Here, $(D^2 + 2D + 1)y = e^{-x} \cdot \log x$ is given

First, find C.F. (y_C), here $r(x) = e^{-x} \cdot \log x$

$$\therefore D^2 + 2D + 1 = 0$$

$$(D+1)^2 = 0$$

$$D = -1, -1 \text{ (-1 two times)}$$

$$\therefore C.F. = (C_1 + C_2 x) e^{-x}$$

$$\therefore y_C = C.F. = C_1 e^{-x} + C_2 x \cdot e^{-x}$$

Now, we know that the general form of C.F.

$$C.F. = C_1 y_1 + C_2 y_2$$

∴

$$y_1 = e^{-x} \text{ and } y_2 = x \cdot e^{-x}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-x} & x e^{-x} \\ -e^{-x} & e^{-x} - x e^{-x} \end{vmatrix} = e^{-2x} - x e^{-2x} + x e^{-2x} = e^{-2x}$$

$$W_1 = -y_2 = -x \cdot e^{-x}$$

$$W_2 = y_1 = e^{-x}$$

Now, for particular integral the formula of variation of parameter is,

$$\text{P.I.} = y_P = y_1 \int \frac{W_1}{W} \cdot r(x) dx + y_2 \int \frac{W_2}{W} \cdot r(x) dx$$

∴

$$\text{P.I.} = y_P = e^{-x} \int \frac{-x \cdot e^{-x}}{e^{-2x}} \cdot e^{-x} \log x dx + x \cdot e^{-x} \int \frac{e^{-x}}{e^{-2x}} \cdot e^{-x} \log x dx$$

$$= -e^{-x} \int \underset{v}{x} \cdot \underset{y}{\log x} dx + x \cdot e^{-x} \int \log x dx$$

⇒ By integration of part we evaluate $\int x \cdot \log x dx$

$$= -e^{-x} \left[\log x \cdot \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} dx \right] + x \cdot e^{-x} [x \log x - x] + C$$

$$= -e^{-x} \left[\log x \cdot \frac{x^2}{2} - \frac{x^2}{4} \right] + x \cdot e^{-x} [x \log x - x] + C$$

$$= -\frac{1}{2} \log x \cdot e^{-x} \cdot x^2 + \frac{1}{4} e^{-x} \cdot x^2 + x^2 e^{-x} \log x - x^2 e^{-x} + C$$

$$y_P = \frac{1}{2} x^2 e^{-x} \log x - \frac{3}{4} x^2 e^{-x} + C$$

∴

$$y = \text{C.F.} + \text{P.I.} = y_C + y_P$$

Ex. 3.5.2 : Solve by method of variation of parameters, $y'' + a^2 y = \tan ax$.

GTU : Summer-18, AEM, Marks 7, Summer-19, Maths-3, Marks 4, Summer-20,21 Maths-2, Marks 4, Winter-23, Maths-3, Marks 7, (a = 2)

Sol. : Here, $(D^2 + a^2)y = \tan ax$ is given

∴ The A.E. is $D^2 + a^2 = 0$

$$D^2 = -a^2 \Rightarrow D = \pm ia$$

∴

$$\text{C.F.} = y_P = C_1 \cos ax + C_2 \sin ax$$

∴

$$y_1 = \cos ax \text{ and } y_2 = \sin ax$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos ax & \sin ax \\ -a \sin ax & a \cos ax \end{vmatrix} = a \cos^2 ax + a \sin^2 ax = a \neq 0$$

$$W_1 = -y_2 = -\sin ax$$

$$W_2 = y_1 = \cos ax$$

Now,

$$y_p = y_1 \int \frac{W_1}{W} \cdot r(x) dx + y_2 \int \frac{W_2}{W} \cdot r(x) dx$$

Where $r(x) = \tan ax$

$$\begin{aligned} \therefore y_p &= \cos ax \int \frac{-\sin ax}{a} \cdot \tan ax dx + \sin ax \int \frac{\cos ax}{a} \cdot \tan ax dx \\ &= -\frac{\cos ax}{a} \int \frac{\sin^2 ax}{\cos ax} dx + \frac{\sin ax}{a} \int \sin ax dx \\ &= -\frac{\cos ax}{a} \int \frac{1 - \cos^2 ax}{\cos ax} dx + \frac{\sin ax}{a} \left(\frac{-\cos ax}{a} \right) \\ &= -\frac{\cos ax}{a} \int \sec ax - \cos ax dx - \frac{\sin ax \cdot \cos ax}{a^2} \\ &= -\frac{\cos ax}{a} \left[\frac{\log |\sec ax + \tan ax|}{a} - \frac{\sin ax}{a} \right] - \frac{\sin ax \cdot \cos ax}{a^2} \\ &= \frac{-\cos ax}{a^2} \cdot \log |\sec ax + \tan ax| + \frac{\cos ax \cdot \sin ax}{a^2} - \frac{\sin ax \cdot \cos ax}{a^2} \end{aligned}$$

$$\therefore \text{P.I.} = y_p = -\frac{\cos ax}{a^2} \log |\sec ax + \tan ax|$$

$$\therefore y = \text{C.F.} + \text{P.I.}$$

$$y = C_1 \cos ax + C_2 \sin ax - \frac{\cos ax}{a^2} \log |\sec ax + \tan ax|$$

Ex. 3.5.3 : Using the method of variation of parameter,

Solve $y'' - 4y' + 4y = \frac{e^{2x}}{x}$.

GTU : Summer-20, Maths-3, Marks 4, Winter-22, AEM, Marks 7

Sol. : Here,

$$y'' - 4y' + 4y = \frac{e^{2x}}{x} \text{ is given}$$

$$\therefore \text{A.E.} \quad D^2 - 4D + 4 = 0$$

$$\therefore (D-2)^2 = 0$$

$$D = 2, 2$$

$$\therefore \text{C.F.} = y_c = e^{2x}(C_1 + C_2 x)$$

$$y_c = C_1 e^{2x} + C_2 x e^{2x}$$

On comparing with C.F. = $C_1 y_1 + C_2 y_2$ we write $y_1 = e^{2x}$ and $y_2 = x e^{2x}$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{2x} & x e^{2x} \\ 2e^{2x} & e^{2x} + 2x e^{2x} \end{vmatrix}$$

$$= e^{4x} + 2x e^{4x} - 2x e^{4x} = e^{4x} \neq 0$$

$$W_1 = -y_2 = -x e^{2x}$$

$$W_2 = y_1 = e^{2x}$$

∴ Method of V.P.

$$y_P = y_1 \int \frac{W_1}{W} \cdot r(x) dx + y_2 \int \frac{W_2}{W} \cdot r(x) dx$$

Where, $r(x) = \frac{e^{2x}}{x}$

$$y_P = e^{2x} \cdot \int \frac{-x \cdot e^{2x}}{e^{4x}} \cdot \frac{e^{2x}}{x} dx + x e^{2x} \int \frac{e^{2x}}{e^{4x}} \cdot \frac{e^{2x}}{x} dx$$

$$= -e^{2x} \int 1 dx + x e^{2x} \int \frac{1}{x} dx$$

$$\therefore y_P = -e^{2x} \cdot x + x \cdot e^{2x} \log x$$

$$\therefore y = y_C + y_P \text{ (Final Answer)}$$

$$\therefore y = e^{2x}(C_1 + C_2 x) + x e^{2x} \log x - x e^{2x}$$

Ex. 3.5.4 : Solve by method of variation of parameters, $\frac{d^2 y}{dx^2} + y = \sec x$.

GTU : Summer-21, Maths-3, Marks 3.5, Summer-20, AEM, Marks 7

Sol. : Here, $y'' + y = \sec x$ is given

A.E. $D^2 + 1 = 0$

$$D^2 = -1$$

$$\Rightarrow D = \pm i$$

$$\therefore \text{C.F.} = C_1 \cos x + C_2 \sin x$$

$$\therefore y_1 = \cos x \text{ and } y_2 = \sin x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$= \cos^2 x + \sin^2 x = 1 \neq 0$$

Now $W_1 = -y_2 = -\sin x$

$$W_2 = y_1 = \cos x$$

$$\therefore y_P = y_1 \int \frac{W_1}{W} \cdot r(x) dx + y_2 \int \frac{W_2}{W} \cdot r(x) dx$$

Where, $r(x) = \sec x$

$$y_P = \cos x \int \frac{-\sin x}{1} \cdot \sec x dx + \sin x \int \frac{\cos x}{1} \cdot \sec x dx$$

$$\therefore y_P = -\cos x \log |\sec x| + \sin x \cdot x$$

$$\therefore y = y_C + y_P$$

$$= C_1 \cos x + C_2 \sin x - \cos x \cdot \log |\sec x| + x \cdot \sin x$$

Ex. 3.5.5 : Using the method of variation of parameters, solve $y'' - 3y' + 2y = e^x$.

GTU : Summer-20, AEM, Marks 3, Summer-22, Maths 3, Marks 4, Summer-24, Maths 3, Marks 4

Sol. : Here, $y'' - 3y' + 2y = e^x$ is given

A.E. $D^2 - 3D + 2 = 0$

$$(D - 2)(D - 1) = 0$$

$$\therefore D = 2 \text{ and } D = 1$$

$$\therefore y_C = \text{C.F.} = C_1 e^x + C_2 e^{2x}$$

Comparing C.F. = $C_1 y_1 + C_2 y_2$

$$\therefore y_1 = e^x \text{ and } y_2 = e^{2x}$$

$$W = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = 2e^{3x} - e^{3x} = e^{3x} \neq 0$$

$$W_1 = -y_2 = -e^{2x}$$

$$W_2 = y_1 = e^x$$

Now, $y_1 = y_1 \int \frac{W_1}{W} \cdot r(x) dx + y_2 \int \frac{W_2}{W} \cdot r(x) dx$

Where, $r(x) = e^x$

$$\therefore y_1 = e^x \int \frac{-e^{2x}}{e^{3x}} e^x dx + e^{2x} \int \frac{e^x}{e^{3x}} \cdot e^x dx$$

$$\begin{aligned} \therefore y_P &= -e^x \int 1 dx + e^{2x} \int e^{-x} dx \\ &= -e^x \cdot x + e^{2x} \left(\frac{e^{-x}}{-1} \right) = -e^x \cdot x - e^x \\ &= -e^x(x+1) \end{aligned}$$

$$\therefore y = y_C + y_P = C_1 e^x + C_2 e^{2x} - e^x(x+1)$$

Ex. 3.5.6 : Using the method of variation of parameter, solve $y'' + 9y = \sec 3x$.

GTU : Summer-19, Winter-21, Maths-2, Marks 7 [9 replace with 4], Summer-23, Maths-3, Marks 7, Summer-23, AEM, Marks 4

Sol. : Here, $y'' + 9y = \sec 3x$ is given

A.E. $D^2 + 9 = 0$

$$D^2 = -9 \Rightarrow D = \pm 3i$$

$$y_C = C_1 \cos 3x + C_2 \sin 3x$$

Take, $y_1 = \cos 3x$ and $y_2 = \sin 3x$

$$W = \begin{vmatrix} \cos 3x & \sin 3x \\ -3 \sin 3x & 3 \cos 3x \end{vmatrix} = 3 \cos^2 3x + 3 \sin^2 3x = 3 \neq 0$$

$$W_1 = -y_2 = -\sin 3x$$

$$W_2 = y_1 = \cos 3x$$

Using V.P. method, find y_P

$\therefore$

$$y_P = y_1 \int \frac{W_1}{W} \cdot r(x) dx + y_2 \int \frac{W_2}{W} \cdot r(x) dx$$

Where,

$$r(x) = \sec 3x$$

$$\begin{aligned} y_P &= \cos 3x \int \frac{-\sin 3x}{3} \cdot \sec 3x dx + \sin 3x \int \frac{\cos 3x}{3} \cdot \sec 3x dx \\ &= -\frac{\cos 3x \log |\sec 3x|}{3} + \frac{\sin 3x}{3} \cdot x \end{aligned}$$

$\therefore$

$$y = y_C + y_P = C_1 \cos 3x + C_2 \sin 3x - \frac{\cos 3x}{9} \log |\sec 3x| + \frac{\sin 3x}{3} \cdot x$$

Ex. 3.5.7 : Solve

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = \frac{e^x}{1+e^x} \text{ using method of variation of parameter.}$$

GTU : Winter-18, Maths-3, Marks 4,
Winter-21, Maths-3, Marks 7

Sol. : Here,

$$y'' - 3y' + 2y = \frac{e^x}{1+e^x} \text{ is given}$$

A.E.

$$D^2 - 3D + 2 = 0$$

$$(D - 2)(D - 1) = 0$$

$$D = 1 \text{ and } D = 2$$

$\therefore$

$$\text{C.F.} = y_C = C_1 e^x + C_2 e^{2x}$$

Comparing with

$$\text{C.F.} = C_1 y_1 + C_2 y_2$$

$\therefore$

$$y_1 = e^x \text{ and } y_2 = e^{2x}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$\therefore$

$$W = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = 2e^{3x} - e^{3x} = e^{3x} \neq 0$$

$$\therefore \quad \begin{aligned} W_1 &= -y_2 = -e^{2x} \\ W_2 &= y_1 = e^x \end{aligned}$$

Variation of parameter is used for finding y_P

$$\therefore \quad y_P = y_1 \int \frac{W_1}{W} \cdot r(x) dx + y_2 \int \frac{W_2}{W} \cdot r(x) dx$$

Where, $r(x) = \frac{e^x}{1+e^x}$

$$\begin{aligned} y_P &= e^x \int \frac{-e^{2x}}{e^{3x}} \cdot \frac{e^x}{1+e^x} dx + e^{2x} \int \frac{e^x}{e^{3x}} \cdot \frac{e^x}{1+e^x} dx \\ &= -e^x \int \frac{1}{1+e^x} dx + e^{2x} \int \frac{1}{e^x(1+e^x)} dx \\ &= -e^x \int \frac{e^{-x}}{1+e^{-x}} dx + e^{2x} \int \left(e^{-x} - \frac{e^{-x}}{1+e^{-x}} \right) dx \\ &= -e^x (-\log(1+e^{-x})) + e^{2x} (-e^{-x} + \log(1+e^{-x})) \\ &= e^x \log(1+e^{-x}) - e^x + e^{2x} \log(1+e^{-x}) \end{aligned}$$

$$\therefore \quad \begin{aligned} y &= y_C + y_P \\ y &= C_1 e^x + C_2 e^{2x} + e^x \log(1+e^{-x}) - e^x + e^{2x} \log(1+e^{-x}) \end{aligned}$$

Ex. 3.5.8 : Solve by method of variation of parameter

$$y'' + 4y = \sec 2x.$$

GTU : Winter-19, Maths-3, Marks 7

Sol. : Here, $y'' + 4y = \sec 2x$ is given

A.E. $D^2 + 4 = 0 \Rightarrow D = \pm 2i$

$$\therefore \quad y_C = C_1 \cos 2x + C_2 \sin 2x$$

$$\therefore \quad y_1 = \cos 2x \quad y_2 = \sin 2x$$

$$\begin{aligned} W &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2x & \sin 2x \\ -2 \sin 2x & 2 \cos 2x \end{vmatrix} \\ &= 2 \cos^2 2x + 2 \sin^2 2x = 2 \neq 0 \end{aligned}$$

$$W_1 = -y_2 = -\sin 2x$$

$$W_2 = y_1 = \cos 2x$$

$$\therefore \quad y_P = y_1 \int \frac{W_1}{W} \cdot r(x) dx + y_2 \int \frac{W_2}{W} \cdot r(x) dx$$

Where, $r(x) = \sec 2x$

$$\therefore \quad y_P = \cos 2x \int \frac{-\sin 2x}{2} \cdot \sec 2x dx + \sin 2x \int \frac{\cos 2x}{2} \cdot \sec 2x dx$$

$$y_P = -\frac{\cos 2x}{2} \frac{\log |\sec 2x|}{2} + \frac{\sin 2x}{2} \cdot x$$

$$y = y_C + y_P$$

$$= C_1 \cos 2x + C_2 \sin 2x - \frac{\cos 2x}{4} \log |\sec 2x| + \frac{x \cdot \sin 2x}{2}$$

Ex. 3.5.9 : Using the method of variation of parameters find the general solution of the differential equation

$$y'' + y = \sin x.$$

GTU : Winter-22, Maths-3, Summer-24, Maths-2, Marks 7

Sol. : Here, $y'' + 4y = \sin x$ is given,

A.E.

$$D^2 + 1 = 0$$

$$\Rightarrow D^2 = -1 \Rightarrow D = \pm i$$

$$\therefore y_C = C_1 \cos x + C_2 \sin x$$

Now, we use variation of parameter to find y_P

$$\therefore \text{Let } y_1 = \cos x \text{ and } y_2 = \sin x$$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1 \neq 0$$

$$W_1 = -y_2 = -\sin x \text{ and } W_2 = y_1 = \cos x$$

$$\therefore y_P = y_1 \int \frac{W_1}{W} \cdot r(x) dx + y_2 \int \frac{W_2}{W} \cdot r(x) dx$$

$$\text{Where, } r(x) = \sin x$$

$$\therefore y_P = \cos x \int \frac{-\sin x}{1} \cdot \sin x dx + \sin x \int \frac{\cos x}{1} \cdot \sin x dx$$

$$= -\cos x \int \sin^2 x dx + \frac{\sin x}{2} \int \sin 2x dx$$

$$= -\cos x \int \frac{1 - \cos 2x}{2} dx + \frac{\sin x}{2} \left(\frac{-\cos 2x}{2} \right)$$

$$= -\frac{\cos x}{2} \left(x - \frac{\sin 2x}{2} \right) - \frac{\sin x \cdot \cos 2x}{4}$$

$$y_P = -\frac{\cos x \cdot x}{2} + \frac{\cos x \cdot \sin 2x}{4} - \frac{\sin x \cdot \cos 2x}{4}$$

Ex. 3.5.10 : Solve by method of variation of parameters

$$\left(\frac{d^2 y}{dx^2} + 9y \right) = \frac{1}{1 + \sin 3x}$$

GTU : Summer-19, AEM, Marks 7

Sol. : Here,

$$y'' + 9y = \frac{1}{1 + \sin 3x} \text{ is given}$$

A.E.

$$D^2 + 9 = 0$$

∴

$$D^2 = -9 \Rightarrow D = \pm 3i$$

$$y_C = C_1 \cos 3x + C_2 \sin 3x$$

Comparing with C.F. = $C_1 y_1 + C_2 y_2$

∴

$$y_1 = \cos 3x \quad y_2 = \sin 3x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 3x & \sin 3x \\ -3 \sin 3x & 3 \cos 3x \end{vmatrix} = 3 \cos^2 3x + 3 \sin^2 3x = 3 \neq 0$$

$$W_1 = -y_2 = -\sin 3x$$

$$W_2 = y_1 = \cos 3x$$

∴

$$y_P = y_1 \int \frac{W_1}{W} \cdot r(x) dx + y_2 \int \frac{W_2}{W} \cdot r(x) dx$$

Where,

$$r(x) = \frac{1}{1 + \sin 3x}$$

$$y_P = \cos 3x \int \frac{-\sin 3x}{3} \cdot \frac{1}{1 + \sin 3x} dx + \sin 3x \int \frac{\cos 3x}{3} \cdot \frac{1}{1 + \sin 3x} dx$$

$$y_P = -\frac{\cos 3x}{3} \int \frac{\sin 3x}{1 + \sin 3x} dx + \frac{\sin 3x}{3} \int \frac{\cos 3x}{1 + \sin 3x} dx$$

∴

$$y_P = -\frac{\cos 3x}{3} \int 1 - \frac{1}{1 + \sin 3x} dx + \frac{\sin 3x}{9} \int \frac{3 \cdot \cos 3x}{1 + \sin 3x} dx \quad \left(\because \int \frac{f'(x)}{f(x)} dx = \log f(x) \right)$$

$$y_P = -\frac{\cos 3x}{3} \left(x - \frac{2}{3 + 3 \tan\left(\frac{3x}{2}\right)} \right) + \frac{\sin 3x}{9} (\log(1 + \sin 3x))$$

∴

$$y = y_C + y_P = C_1 \cos 3x + C_2 \sin 3x - \frac{\cos 3x}{3} \cdot x + \frac{2}{9} \cdot \frac{\cos 3x}{1 + \tan\left(\frac{3x}{2}\right)} + \frac{\sin 3x}{9} (\log(1 + \sin 3x))$$

Ex. 3.5.11 : Using the method of variation of parameter find the general solution of $(D^2 - 2D + 1)y = 3x^{3/2}e^x$.

GTU : Winter-19, Summer-24, AEM, Marks 7

Sol. : Here, $(D^2 - 2D + 1)y = 3x^{3/2}e^x$ is given

∴ A.E.

$$D^2 - 2D + 1 = 0$$

$$(D - 1)^2 = 0$$

∴

$$D = 1, 1$$

∴

$$y_C = e^x(C_1 + C_2 x) = C_1 e^x + C_2 x e^x$$

Now,

$$y_1 = e^x \quad y_2 = x e^x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & x e^x \\ e^x & e^x + x e^x \end{vmatrix} = e^{2x} + x e^{2x} - x e^{2x} = e^{2x} \neq 0$$

$$W_1 = -y_2 = -x e^x$$

$$W_2 = y_1 = e^x$$

$$\therefore y_P = y_1 \int \frac{W_1}{W} \cdot r(x) dx + y_2 \int \frac{W_2}{W} \cdot r(x) dx$$

Where,

$$r(x) = 3x^{3/2}e^x$$

$$\therefore y_P = e^x \int \frac{-xe^x}{e^{2x}} \cdot 3x^{3/2}e^x dx + x e^x \int \frac{e^x}{e^{2x}} \cdot 3x^{3/2}e^x dx$$

$$\therefore y_P = -3e^x \int x^{5/2} dx + 3xe^x \int x^{3/2} dx = -3e^x \left[\frac{x^{7/2}}{7/2} \right] + 3xe^x \left[\frac{x^{5/2}}{5/2} \right]$$

$$= -\frac{6}{7}e^x x^{7/2} + \frac{6}{5}e^x x^{7/2} = \frac{12}{35}e^x \cdot x^{7/2}$$

$$\therefore y = y_C + y_P$$

$$\therefore y = e^x (C_1 + C_2 x) + \frac{12}{35}e^x \cdot x^{7/2}$$

Ex. 3.5.12 : Solve using - method of variation of parameters $y'' + 2y' + y = e^{-x} \cos x$.

GTU : Winter-18, AEM, Marks 7

Sol. : Here, $y'' + 2y' + y = e^{-x} \cos x$ is given

$$\text{A.E.} \quad D^2 + 2D + 1 = 0$$

$$\therefore (D+1)^2 = 0$$

$$\therefore D = -1, -1$$

$$\therefore y_C = e^{-x}(C_1 + C_2 x)$$

Comparing this with C.F. = $C_1 y_1 + C_2 y_2$

$$\therefore y_1 = e^{-x} \text{ and } y_2 = x e^{-x}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-x} & x e^{-x} \\ -e^{-x} & e^{-x} - x e^{-x} \end{vmatrix}$$

$$= e^{-2x} - x e^{-2x} + x e^{-2x} = e^{-2x} \neq 0$$

$$W_1 = -y_2 = -x e^{-x}$$

$$W_2 = y_1 = e^{-x}$$

$$\therefore y_P = y_1 \int \frac{W_1}{W} \cdot r(x) dx + y_2 \int \frac{W_2}{W} r(x) dx$$

where,

$$r(x) = e^{-x} \cos x$$

$$\therefore y_P = e^{-x} \int \frac{-x e^{-x}}{e^{-2x}} \cdot e^{-x} \cos x dx + x e^{-x} \int \frac{e^{-x}}{e^{-2x}} e^{-x} \cos x dx$$

$$\therefore y_P = -e^{-x} \int x \cos x dx + x e^{-x} \int \cos x dx = -e^{-x} [x \sin x - (-\cos x)] + x e^{-x} (\sin x)$$

$$= -x e^{-x} \sin x - e^{-x} \cos x + x e^{-x} \sin x$$

$$\therefore y_P = -e^{-x} \cos x$$

$$\therefore y = y_C + y_P = e^{-x}(C_1 + C_2 x) - e^{-x} \cos x$$

Ex. 3.5.13 : Solve the differential equation $y''' + y' = \operatorname{cosec} x$ by the method of variation of parameter.

GTU : Summer-12, Maths-3, Marks 7

Sol. : Here, $y''' + y' = \operatorname{cosec} x$ is given

A.E. $D^3 + D = 0$

$$D(D^2 + 1) = 0$$

$$D = 0 \text{ and } D^2 + 1 = 0$$

$$D^2 = -1$$

$$\therefore D = \pm i$$

$$\therefore \text{C.F.} = y_C = C_1 + C_2 \cos x + C_3 \sin x$$

Comparing with $y_C = C_1 y_1 + C_2 y_2 + C_3 y_3$

$$\therefore y_1 = 1, y_2 = \cos x, y_3 = \sin x$$

$$\therefore W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 1 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ 0 & -\cos x & -\sin x \end{vmatrix}$$

$$= \sin^2 x + \cos^2 x = 1 \neq 0$$

Now,

$$W_1 = \begin{vmatrix} 0 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ 1 & -\cos x & -\sin x \end{vmatrix} = \sin^2 x + \cos^2 x = 1 \neq 0$$

$$W_2 = \begin{vmatrix} 1 & 0 & \sin x \\ 0 & 0 & \cos x \\ 0 & 1 & -\sin x \end{vmatrix} = -\cos x$$

$$W_3 = \begin{vmatrix} 1 & \cos x & 0 \\ 0 & -\sin x & 0 \\ 0 & -\cos x & 1 \end{vmatrix} = -\sin x$$

$$y_P = y_1 \int \frac{W_1}{W} r(x) dx + y_2 \int \frac{W_2}{W} r(x) dx + y_3 \int \frac{W_3}{W} r(x) dx$$

$$= 1 \int \frac{1}{1} \cdot \operatorname{cosec} x dx + \cos x \int \frac{-\cos x}{1} \cdot \operatorname{cosec} x dx + \sin x \int \frac{-\sin x}{1} \cdot \operatorname{cosec} x dx$$

$$= \log |\operatorname{cosec} x - \cot x| - \cos x \log |\sin x| - \sin x \cdot x$$

$$y_P = C_1 + C_2 \cos x + C_3 \sin x + \log |\operatorname{cosec} x - \cot x| - \cos x \log |\sin x| - x \cdot \sin x$$

$$y = y_C + y_P$$

$$= C_1 + C_2 \cos x + C_3 \sin x + \log |\operatorname{cosec} x - \cot x| - \cos x \log |\sin x| - x \sin x$$

Ex. 3.5.14 : Apply the method of variation of parameter to solve

$$y''' - 9y'' + 27y' - 27y = \frac{e^{3x}}{x^2}$$

Sol :

Here, $y''' - 9y'' + 27y' - 27y = \frac{e^{3x}}{x^2}$

A.E.

$$D^3 - 9D^2 + 27D - 27 = 0$$

$$(D-3)^3 = 0$$

$$D = 3, 3, 3$$

$$\text{C.F.} = y_C = e^{3x}(C_1 + C_2x + C_3x^2)$$

$$\text{C.F.} = C_1e^{3x} + C_2xe^{3x} + C_3x^2e^{3x}$$

$$\text{Comparing with C.F.} = C_1y_1 + C_2y_2 + C_3y_3$$

$$\therefore y_1 = e^{3x}, y_2 = x \cdot e^{3x}, y_3 = x^2e^{3x}$$

$$\therefore W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} e^{3x} & xe^{3x} & x^2e^{3x} \\ 3e^{3x} & e^{3x} + 3xe^{3x} & (3x^2 + 2)e^{3x}x \\ 9e^{3x} & 9xe^{3x} + 6e^{3x} & (9x^2 + 12x + 2)e^{3x} \end{vmatrix}$$

$$= e^{9x} \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & 2x \\ 0 & 6 & 12x + 2 \end{vmatrix} = 2 \cdot e^{9x}$$

$$W_1 = \begin{vmatrix} 0 & xe^{3x} & x^2e^{3x} \\ 0 & e^{3x} + 3xe^{3x} & (3x^2 + 2)e^{3x}x \\ 1 & 9xe^{3x} + 6e^{3x} & (9x^2 + 12x + 2)e^{3x} \end{vmatrix} = x^2e^{6x}$$

$$W_2 = \begin{vmatrix} e^{3x} & 0 & x^2e^{3x} \\ 3e^{3x} & 0 & (3x^2 + 2)e^{3x}x \\ 9e^{3x} & 1 & (9x^2 + 12x + 2)e^{3x} \end{vmatrix} = -2xe^{6x}$$

$$W_3 = \begin{vmatrix} e^{3x} & xe^{3x} & 0 \\ 3e^{3x} & e^{3x} + 3xe^{3x} & 0 \\ 9e^{3x} & 9xe^{3x} + 6e^{3x} & 1 \end{vmatrix} = e^{6x}$$

$$\begin{aligned} \therefore y_P &= y_1 \int \frac{W_1}{W} r(x) dx + y_2 \int \frac{W_2}{W} r(x) dx + y_3 \int \frac{W_3}{W} r(x) dx \\ &= e^{3x} \int \frac{x^2e^{6x}}{2e^{9x}} \cdot \frac{e^{3x}}{x^2} dx + x \cdot e^{3x} \int \frac{-2xe^{6x}}{2e^{9x}} \cdot \frac{e^{3x}}{x^2} dx + x^2e^{3x} \int \frac{e^{6x}}{2e^{9x}} \cdot \frac{e^{3x}}{x^2} dx \\ &= \frac{e^{3x}}{2} \int 1 dx + x \cdot e^{3x} \int \frac{-1}{x} dx + \frac{x^2e^{3x}}{2} \int \frac{1}{x^2} dx \end{aligned}$$

$$y_P = \frac{e^{3x}}{2} x - xe^{3x} \log x - \frac{x^2e^{3x}}{2} \left(-\frac{1}{x} \right)$$

$$\therefore y_P = -xe^{3x} \log x$$

∴

$$y = y_C + y_P = C_1 e^{3x} + C_2 x e^{3x} + C_3 x^2 e^{3x} - x e^{3x} \log x$$

Ex. 3.5.15 : Solve $y'' + y = \operatorname{cosec} x$ using variation of parameter method.

GTU : Winter-21, AEM, Marks 7

Sol. : Here, $y'' + y = \operatorname{cosec} x$ is given

A.E.

$$D^2 + 1 = 0$$

$$D^2 = -1$$

∴

$$D = \pm i$$

∴

$$y_C = C_1 \cos x + C_2 \sin x$$

Compare with C.F. = $C_1 y_1 + C_2 y_2$

∴

$$y_1 = \cos x \quad y_2 = \sin x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1 \neq 0$$

$$W_1 = -y_2 = -\sin x$$

$$W_2 = y_1 = \cos x$$

$$y_P = y_1 \int \frac{W_1}{W} r(x) dx + y_2 \int \frac{W_2}{W} r(x) dx$$

where,

$$r(x) = \operatorname{cosec} x$$

∴

$$y_P = \cos x \int \frac{\sin x}{1} \cdot \operatorname{cosec} x dx + \sin x \int \frac{\cos x}{1} \cdot \operatorname{cosec} x dx$$

$$= -\cos x \int 1 dx + \sin x \int \cot x dx$$

$$= -\cos x \cdot x + \sin x \log |\sin x|$$

$$y = y_C + y_P$$

$$y = C_1 \cos x + C_2 \sin x - x \cdot \cos x + \sin x \log |\sin x|$$

Ex. 3.5.16 : Using method of variation of parameter solve $y'' - 2y' + y = x e^x \sin x$.

GTU : Winter-23, Maths-2, Marks 7

Sol. : Here, $y'' - 2y' + y = x e^x \sin x$ is given

A.E.

$$D^2 - 2D + 1 = 0$$

$$(D-1)^2 = 0$$

$$D = 1, 1$$

∴

$$y_C = e^x (C_1 + C_2 x)$$

∴

$$y_C = C_1 e^x + C_2 x e^x$$

Let,

$$y_1 = e^x \quad y_2 = x e^x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & x e^x \\ e^x & e^x + x e^x \end{vmatrix}$$

$$= e^{2x} + x e^{2x} - x e^{2x} = e^{2x} \neq 0$$

$$W_1 = -y_2 = -x e^x$$

$$W_2 = y_1 = e^x$$

$$y_P = y_1 \int \frac{W_1}{W} \cdot r(x) dx + y_2 \int \frac{W_2}{W} \cdot r(x) dx$$

$$y_P = e^x \int \frac{-x e^x}{e^{2x}} \cdot x e^x \sin x dx + x e^x \int \frac{e^x}{e^{2x}} x e^x \sin x dx$$

$$= -e^x \int x^2 \sin x dx + x e^x \int x \sin x dx$$

$$= -e^x [(x^2)(-\cos x) - (2x)(-\sin x) + (2)(\cos x)] + x e^x [(x)(-\cos x) - (1)(-\sin x)]$$

$$= e^x x^2 \cos x - 2e^x x \sin x - 2e^x \cos x - x^2 e^x \cos x + x e^x \sin x$$

$$= -e^x x \sin x - 2e^x \cos x$$

$$= -e^x (x \sin x + 2 \cos x)$$

$$y = y_C + y_P$$

$$y = C_1 e^x + C_2 x e^x - e^x (x \sin x + 2 \cos x)$$

Ex. 3.5.17 : Solve the differential equation : $(D^2 - 2D + 1)y = e^x$ using method of variation of parameters.

Sol. : Auxiliary equation is : $(m^2 - 2m + 1) = 0$

$$\Rightarrow m = 1, 1$$

$$\text{C.F.} = (C_1 + C_2 x) e^x = C_1 e^x + C_2 x e^x = C_1 y_1 + C_2 y_2$$

$$\therefore y_1 = e^x \text{ and } y_2 = x e^x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & x e^x \\ e^x & x e^x + e^x \end{vmatrix} = e^{2x}$$

$$u_1 = -\int \frac{y_2 F(x)}{W} dx = -\int \frac{x e^x e^x}{e^{2x}} dx$$

$$= -\int x dx = -\frac{x^2}{2}$$

$$u_2 = \int \frac{y_1 F(x)}{W} dx = \int \frac{e^x e^x}{e^{2x}} dx = \int 1 dx = x$$

$$\therefore \text{P.I.} = u_1 y_1 + u_2 y_2$$

$$= -\frac{x^2}{2} e^x + x^2 e^x = \frac{x^2}{2} e^x$$

Complete solution is : $y = \text{C.F.} + \text{P.I.}$

$$\Rightarrow y = (C_1 + C_2 x) e^x + \frac{x^2}{2} e^x$$

Ex. 3.5.18 : Solve the differential equation :

$\frac{d^2y}{dx^2} + 4y = x \sin 2x$ using method of variation of parameters.

Sol. :

$$\Rightarrow (D^2 + 4)y = x \sin 2x$$

Auxiliary equation is : $(m^2 + 4) = 0$

$$\Rightarrow m = \pm 2i$$

$$\text{C.F.} = C_1 \cos 2x + C_2 \sin 2x = C_1 y_1 + C_2 y_2$$

$$\therefore y_1 = \cos 2x \quad \text{and} \quad y_2 = \sin 2x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2x & \sin 2x \\ -2 \sin 2x & 2 \cos 2x \end{vmatrix} = 2$$

$$\begin{aligned} u_1 &= -\int \frac{y_2 F(x)}{W} dx = -\frac{1}{2} \int x \sin^2 2x dx = -\frac{1}{4} \int x(1 - \cos 4x) dx \\ &= -\frac{1}{4} \left[\frac{x^2}{2} - \left(x \left(\frac{\sin 4x}{4} \right) - (1) \left(-\frac{\cos 4x}{16} \right) \right) \right] = \left[-\frac{x^2}{8} + \frac{x \sin 4x}{16} + \frac{\cos 4x}{64} \right] \end{aligned}$$

$$\begin{aligned} u_2 &= \int \frac{y_1 F(x)}{W} dx = \frac{1}{2} \int x \sin 2x \cos 2x dx = \frac{1}{4} \int x \sin 4x dx \\ &= \frac{1}{4} \left[x \left(-\frac{\cos 4x}{4} \right) - (1) \left(-\frac{\sin 4x}{16} \right) \right] = \left[-\frac{x \cos 4x}{16} + \frac{\sin 4x}{64} \right] \end{aligned}$$

$$\begin{aligned} \therefore \text{P.I.} &= u_1 y_1 + u_2 y_2 \\ &= \cos 2x \left[-\frac{x^2}{8} + \frac{x \sin 4x}{16} + \frac{\cos 4x}{64} \right] + \sin 2x \left[-\frac{x \cos 4x}{16} + \frac{\sin 4x}{64} \right] \\ &= \frac{x}{16} (\sin 4x \cos 2x - \cos 4x \sin 2x) + \frac{1}{64} (\cos 4x \cos 2x + \sin 4x \sin 2x) - \frac{x^2}{8} \cos 2x \\ &= \frac{x}{16} \sin 2x + \frac{1}{64} \cos 2x - \frac{x^2}{8} \cos 2x \end{aligned}$$

Complete solution is : $y = \text{C.F.} + \text{P.I.}$

$$\Rightarrow y = C_1 \cos 2x + C_2 \sin 2x + \frac{x}{16} \sin 2x + \frac{1}{64} \cos 2x - \frac{x^2}{8} \cos 2x$$

Ex. 3.5.19 : Solve the differential equation : $(D^2 - D - 2)y = e^{(e^x + 3x)}$ using method of variation of parameters.

Sol. : Auxiliary equation is : $(m^2 - m - 2) = 0$

$$\Rightarrow m = -1, 2$$

$$\text{C.F.} = C_1 e^{-x} + C_2 e^{2x} = C_1 y_1 + C_2 y_2$$

$$\therefore y_1 = e^{-x} \quad \text{and} \quad y_2 = e^{2x}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-x} & e^{2x} \\ -e^{-x} & 2e^{2x} \end{vmatrix} = 3e^x$$

$$u_1 = - \int \frac{y_2 F(x)}{W} dx = - \int \frac{e^{2x} e(e^x + 3x)}{3e^x} dx = - \int \frac{e^{2x} e^x e^{3x}}{3e^x} dx$$

$$= - \frac{1}{3} \int e^{4x} e^x dx, \text{ Putting } e^x = t \Rightarrow e^x dx = t dt$$

$$u_1 = - \frac{1}{3} \int t^3 e^t dt = - \frac{1}{3} [(t^3)(e^t) - (3t^2)(e^t) + (6t)(e^t) - (6)(e^t)]$$

$$u_1 = - \frac{e^{e^x}}{3} [e^{3x} - 3e^{2x} + 6e^x - 6]$$

$$u_2 = \int \frac{y_1 F(x)}{W} dx = \int \frac{e^{-x} e(e^x + 3x)}{3e^x} dx = \int \frac{e^{-x} e^x e^{3x}}{3e^x} dx = \frac{1}{3} \int e^x e^x dx = \frac{e^{e^x}}{3}$$

$$\therefore \text{P.I.} = u_1 y_1 + u_2 y_2$$

$$= - \frac{e^{e^x} e^{-x}}{3} [e^{3x} - 3e^{2x} + 6e^x - 6] + \frac{e^{e^x} e^{2x}}{3} = \frac{e^{e^x}}{3} [3e^x - 6 + 6e^{-x}]$$

Complete solution is : $y = \text{C.F.} + \text{P.I.}$

$$y = C_1 e^{-x} + C_2 e^{2x} + \frac{e^{e^x}}{3} [3e^x - 6 + 6e^{-x}]$$

3.6 Euler-Cauchy Equations

GTU : Winter-11,12,17,22,23, Summer-12,18,19,20,22,23,24

Euler Cauchy's Equations :

- A differential equation of the form

$$x^n \frac{dy^n}{dx^n} + C_1 x^{n-1} \frac{dy^{n-1}}{dx^{n-1}} + C_2 x^{n-2} \frac{dy^{n-2}}{dx^{n-2}} + \dots + C_n y = f(x) \quad \dots(3.6.1)$$

- Where, $C_1, C_2, \dots, C_n$ are constant and $f(x)$ is a function of x is called **Euler Cauchy's** linear equation of order n .
- Now equation (3.6.1) can be convert in to linear differential equation with constant coefficient by substituting.

$$x = e^z \quad \text{or} \quad z = \log x$$

Now,

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx}$$

$$z = \log x \Rightarrow \frac{dz}{dx} = \frac{1}{x}$$

$$\therefore \frac{dy}{dx} = \frac{1}{x} \cdot \frac{dy}{dz}$$

$$\therefore x \cdot \frac{dy}{dx} = \frac{dy}{dz} = Dy, \quad \text{where } D = \frac{d}{dz}$$

Similarly, $x^2 \frac{d^2 y}{dx^2} = D(D-1)y$

$$x^3 \frac{d^3 y}{dx^3} = D(D-1)(D-2)y \text{ and so on}$$

- Substituting these value in equation (3.6.1), it can be convert to linear differential equation with constant co-efficient.
- This types of equation solved by the earlier discussed method.

Ex. 3.6.1 : Solve $(x^2 D^2 + 3xD + 4)y = 0$,
 $y(1) = 0$, $y'(1) = 3$.

GTU : Winter-11, Maths-3, Marks 4
Summer-22, Maths-2, Marks 4

Sol. : Here, $(x^2 D^2 - 3xD + 4)y = 0$ is given

Let, $x = e^z \Rightarrow z = \log x$

$$\therefore x \cdot \frac{dy}{dx} = Dy \text{ and } x^2 \frac{d^2 y}{dx^2} = D(D-1)y$$

Where, $D = \frac{d}{dz}$

$\therefore$ Given equation can be express as,

$$(D(D-1) - 3D + 4)y = 0$$

$$\therefore (D^2 - 4D + 4)y = 0$$

$$\therefore \text{A.E. } D^2 - 4D + 4 = 0$$

$$(D-2)^2 = 0$$

$$\therefore D = 2, 2$$

$$\text{C.F. } y = e^{2z}(C_1 + C_2 z)$$

$$\therefore y = x^2(C_1 + C_2 \log x)$$

Now, initial condition is given,

$$y(1) = 0 \Rightarrow y = 0 \text{ when } x = 1$$

$$0 = 1(C_1 + C_2 \log 1)$$

$$C_1 = 0$$

Take derivative of y.

$$y' = 2x(C_1 + C_2 \log x) + x^2 \left(C_2 + \frac{1}{x} \right)$$

Here, $y'(1) = 3$ is given

$$\Rightarrow y' = 3 \text{ when } x = 1$$

$$\therefore 3 = 2C_1 + C_2$$

$$\therefore C_2 = 3$$

$$\therefore y = 3x^2 \log x$$

Ex. 3.6.2 : Solve $(4x^2 D^2 + 16xD + 9)y = 0$.

GTU : Winter-23, AEM, Marks 7

Sol. : Here, $(4x^2 D^2 + 16xD + 9)y = 0$ is given

This is Euler Cauchy's equation.

$$\therefore \text{Let, } x = e^z \Rightarrow z = \log x$$

$$x \cdot \frac{dy}{dx} = Dy$$

$$x^2 \cdot \frac{d^2 y}{dx^2} = D(D-1)y \text{ where, } D = \frac{d}{dz}$$

$$\therefore (4D(D-1) + 16D + 9)y = 0$$

$$\therefore (4D^2 + 12D + 9)y = 0$$

$$\therefore \text{A.E. } 4D^2 + 12D + 9 = 0$$

$$4D^2 + 6D + 6D + 9 = 0$$

$$2D(2D + 3) + 3(2D + 3) = 0$$

$$\therefore (2D + 3)^2 = 0$$

$$\therefore 2D = -3 \Rightarrow D = \frac{-3}{2}$$

$$\therefore y_C = (C_1 + C_2 z)e^{-3/2z}$$

$$\therefore y = (C_1 + C_2 \log z)x^{-3/2}$$

Ex. 3.6.3 : $x^2 y'' - 20y = 0$.

GTU : Summer-24, Mathe-2, Marks 3

Sol. : Here, $x^2 y'' - 20y = 0$

This is a Euler's Cauchy equation,

$$\therefore x = e^z \Rightarrow z = \log x$$

$$x \frac{dy}{dx} = Dy, \quad x^2 \frac{d^2 y}{dx^2} = D(D-1)y$$

$$\therefore (D(D-1) - 20)y = 0$$

$$\therefore (D^2 - D - 20)y = 0$$

A.E. $D^2 - D - 20 = 0$

$$D^2 - 5D + 4D - 20 = 0$$

$$D(D - 5) + 4(D - 5) = 0$$

$$(D + 4)(D - 5) = 0$$

$$D = 4, 5$$

$$y = C_1 e^{-4z} + C_2 e^{5z}$$

$$y = C_1 \frac{1}{x^4} + C_2 x^5$$

Ex. 3.6.4 : Solve $(x^2 D^2 - 7xD + 12)y = x^2$.

GTU : Winter-22, Maths-3, Marks 7
Winter-22, Maths-2, Marks 4

Sol. : Here, $(x^2 D^2 - 7xD + 12)y = x^2$ is given

This is Euler's Cauchy equation,

$$\therefore x = e^z \Rightarrow z = \log x, \text{ where, } D = \frac{d}{dx}$$

$$\therefore x \cdot \frac{dy}{dx} = Dy$$

$$\therefore x^2 \frac{d^2 y}{dx^2} = D(D - 1)y$$

$\therefore$ Given equation,

$$(D(D - 1) - 7D + 12)y = e^{2z} \quad \dots(1)$$

A.E. $D(D - 1) - 7D + 12 = 0$

$$\therefore D^2 - 8D + 12 = 0$$

$$\therefore (D - 6)(D - 2) = 0$$

$$\therefore D = 6 \text{ and } D = 2$$

$$y_C = C_1 e^{6z} + C_2 e^{2z}$$

Let us consider trial solution,

$$y_P = Ce^{2z}$$

Here, e^{2z} appears once in y_C .

$\therefore$ We multiply the corresponding term by z .

$$\therefore y_P = Cze^{2z}$$

$$y'_P = Ce^{2z} + 2Ce^{2z}z$$

$$y''_P = 2Ce^{2z} + 2Ce^{2z} + 4Cze^{2z}$$

$$\therefore y''_P = 4Ce^{2z} + 4Cze^{2z}$$

Now, Put y_P , y'_P and y''_P in given equation (1)

$$\therefore (D^2 - 8D + 12)y = e^{2z}$$

$$\therefore y''_P - 8y'_P + 12y_P = e^{2z}$$

$$\therefore 4Ce^{2z} + 4Cze^{2z} - 8Ce^{2z} - 16Cze^{2z} + 12Cze^{2z} = e^{2z}$$

$$-4Ce^{2z} = e^{2z}$$

$$\Rightarrow -4C = 1 \Rightarrow C = -\frac{1}{4}$$

$$\therefore y_P = -\frac{1}{4}ze^{2z}$$

$$y = y_C + y_P = C_1 e^{6z} + C_2 e^{2z} - \frac{1}{4}ze^{2z}$$

$$\therefore y = C_1 x^6 + C_2 x^2 - \frac{1}{4} \log x \cdot x^2$$

Ex. 3.6.5 : Solve $x^2 y'' - xy' + y = \sin(\log x)$.

GTU : Summer-17, Maths-3, Marks 7

Sol. : Here, $x^2 y'' - xy' + y = \sin(\log x)$ is given

This is Euler's Cauchy equation,

Let us consider $x = e^z \Rightarrow z = \log x$

Now, $x \cdot y' = Dy, \quad D = \frac{d}{dz}$

$$x^2 y'' = D(D - 1)y$$

$\therefore$ We write given equation,

$$(D(D - 1) - D + 1)y = \sin z$$

$$\therefore (D^2 - 2D + 1)y = \sin z$$

$\therefore$ A.E. $D^2 - 2D + 1 = 0$

$$\therefore (D - 1)^2 = 0$$

$$D = 1, 1$$

$$\text{C.F.} = (C_1 + C_2 z)e^z$$

Here, we use U.C. for finding P.I.

Let us consider trial solutions y_P

$$\therefore y_P = C_1 \cos z + C_2 \sin z$$

$$y'_P = -C_1 \sin z + C_2 \cos z$$

$$y_P'' = -C_1 \cos z - C_2 \sin z$$

Put y_P , y_P' and y_P'' in given equation,

$$\therefore y_P'' - 2y_P' + y_P = \sin z$$

$$-C_1 \cos z - C_2 \sin z + 2C_1 \sin z - 2C_2 \cos z + C_1 \cos z + C_2 \sin z = \sin z$$

$$2C_1 \sin z - 2C_2 \cos z = \sin z$$

Comparing coefficient both sides,

$$\therefore 2C_1 = 1 \Rightarrow C_1 = \frac{1}{2}$$

$$-2C_2 = 0 \Rightarrow C_2 = 0$$

$$\therefore y_P = \frac{1}{2} \cos z$$

$$\therefore y = y_C + y_P = (C_1 + C_2 z) e^z + \frac{1}{2} \cos z$$

$$\therefore y = (C_1 + C_2 (\log x)) x + \frac{1}{2} \cos (\log x)$$

Ex. 3.6.6 : Solve $(x^2 D^2 - 3xD + 4)y = x^2$

GTU : Summer-12, Maths-3, Marks 7

Sol. : Here, $(x^2 D^2 - 3xD + 4)y = x^2$

$$\text{Let, } x = e^z \Rightarrow z = \log x$$

$$\text{and } x \cdot \frac{dy}{dx} = Dy, \quad \frac{x^2 d^2 y}{dx^2} = D(D-1)y$$

$$\text{where, } D = \frac{d}{dz}$$

$\therefore$ Given equation can be written as,

$$(D(D-1) - 3D + 4)y = e^{2z}$$

$$\therefore (D^2 - 4D + 4)y = e^{2z}$$

$$\text{Now, A.E. } D^2 - 4D + 4 = 0$$

$$(D-2)^2 = 0$$

$$\therefore \text{C.F.} = (C_1 + C_2 z) e^{2z}$$

Here, we use U.C. method to find P.I.

$\therefore$ Let us consider trial solution y_P

$$y_P = C e^{2z}$$

Here, e^{2z} term appears twice in C.F.

$\therefore$ We multiply z^2 in y_P

$$\therefore y_P = C z^2 e^{2z}$$

$$y'_P = 2Cze^{2z} + 2Cz^2e^{2z}$$

$$y''_P = 2Ce^{2z} + 4Cze^{2z} + 4Cze^{2z} + 4Cz^2e^{2z}$$

$$y''_P = 2Ce^{2z} + 8Cze^{2z} + 4Cz^2e^{2z}$$

Put y_P , y'_P and y''_P in given equation,

$$y''_P - 4y'_P + 4y_P = e^{2z}$$

$$2Ce^{2z} + 8Cze^{2z} + 4Cz^2e^{2z} - 8Cze^{2z} - 8Cz^2e^{2z} + 4Cz^2e^{2z} = e^{2z}$$

$$2Ce^{2z} = e^{2z}$$

$$2C = 1 \Rightarrow C = \frac{1}{2}$$

$$y_P = \frac{1}{2}z^2e^{2z}$$

$$y = y_C + y_P = (C_1 + C_2z)e^{2z} + \frac{1}{2}z^2e^{2z}$$

$$y = (C_1 + C_2 \log x)x^2 + \frac{1}{2}x^2(\log x)^2$$

Ex. 3.6.7 : Solve $x^2y'' - 2xy' + y = \cos(\log x)$

GTU : Summer-20, Maths-2, Marks 4; Summer-23, Maths-2, Marks 3.5

Sol. : Here, $x^2y'' - 2xy' + y = \cos(\log x)$ is given

This is Cauchy's Euler equation,

$$x = e^z \Rightarrow z = \log x$$

$$xy' = Dy \text{ and } x^2y'' = D(D-1)y$$

where,

$$D = \frac{d}{dz}$$

$$(D(D-1) - 2D + 1)y = \cos z$$

$$(D^2 - 3D + 1)y = \cos z$$

$$\text{A.E. } D^2 - 3D + 1 = 0$$

$$a = 1, b = -3, c = 1$$

$$\Delta = b^2 - 4ac = 9 - 4 = 5$$

$$D_1, D_2 = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{3 \pm \sqrt{5}}{2}$$

$$\therefore \text{C.F.} = y_P = C_1 e^{\frac{3+\sqrt{5}}{2}z} + C_2 e^{\frac{3-\sqrt{5}}{2}z}$$

For finding y_P we use U.C. method,

$$y_P = A \cos z + B \sin z$$

$$y'_P = -A \sin z + B \cos z$$

$$y_P'' = -A \cos z - B \sin z$$

Put, y_P , y_P' , y_P'' in given equation.

$$y_P'' - 3y_P' + y_P = \cos z$$

$$-A \cos z - B \sin z + 3A \sin z - 3B \cos z + A \cos z + B \sin z = \cos z$$

$$3A \sin z - 3B \cos z = \cos z$$

Comparing co-efficient both sides

$$3A = 0 \Rightarrow A = 0$$

$$-3B = 1 \Rightarrow B = -\frac{1}{3}$$

$\therefore$

$$y_P = -\frac{1}{3} \sin z$$

$$y = y_C + y_P$$

$$y = C_1 e^{\frac{3+\sqrt{5}}{2}z} + C_2 e^{\frac{3-\sqrt{5}}{2}z} - \frac{1}{3} \sin z$$

$$y = C_1 x^{\frac{3+\sqrt{5}}{2}} + C_2 x^{\frac{3-\sqrt{5}}{2}} - \frac{1}{3} \sin (\log x)$$

Ex. 3.6.8 : Solve $x^3 y''' + x^2 y'' = x^3$.

GTU : Winter-23, Maths-2, Marks 4

Sol. : This is Euler Cauchy's equation,

$$x^3 y''' + x^2 y'' = x^3$$

Let $x = e^z \Rightarrow z = \log x$

and $xy' = Dy$

$$x^2 y'' = D(D-1)y$$

$$x^3 y''' = D(D-1)(D-2)y$$

where, $D = \frac{d}{dz}$

$$(D(D-1)(D-2) + D(D-1))y = e^{3z}$$

$$(D^3 - 2D^2 + D)y = e^{3z}$$

A.E. $D(D-1)(D-2+1) = 0$

$$D(D-1)(D-1) = 0$$

$$D = 0, 1, 1$$

$$y_C = C.F. = C_1 e^z (C_2 + C_3 z)$$

Now, let us consider trial solution y_P

$\therefore y_P = C e^{3z}$

$$y_P' = 3Ce^{3z}, \quad y_P'' = 9Ce^{3z} \quad \text{and} \quad y_P''' = 27Ce^{3z}$$

Put y_P', y_P'' in $(D^3 - 2D^2 + D)y = e^{3z}$

$$\therefore y_P''' - 2y_P'' + y_P' = e^{3z}$$

$$27Ce^{3z} - 18Ce^{3z} + 3Ce^{3z} = e^{3z}$$

$$30Ce^{3z} - 18Ce^{3z} = e^{3z}$$

$$\therefore 12C = 1 \Rightarrow C = \frac{1}{12}$$

$$\therefore y_P = \frac{1}{2}e^{3z}$$

$$\therefore y = y_C + y_P$$

$$y = C_1 + e^z(C_2 + C_3 z) + \frac{1}{12}e^{3z}$$

$$\therefore y = C_1 + x(C_2 + C_3 \log x) + \frac{1}{12}x^3$$

Ex. 3.6.9 : Solve $(x^2 D^2 + xD + 1)y = \log x \cdot \sin(\log x)$

GTU : Summer-18, Maths-3, Marks 4
Winter-23, Maths-3, Marks 7

Sol. : Here, $(x^2 D^2 + xD + 1)y = \log x \cdot \sin(\log x)$ is given

This equation is Cauchy's linear equations,

Let, $x = e^z \Rightarrow z = \log x$

$$\therefore x \frac{dy}{dx} = Dy$$

$$x^2 \frac{d^2 y}{dx^2} = D(D-1)y, \quad \text{where } D = \frac{d}{dz}$$

We write given equation in the form

$$(D(D-1) + D + 1)y = z \sin z$$

For, C.F. $D(D-1) + D + 1 = 0$

$$D^2 - D + D + 1 = 0$$

$$\therefore D^2 + 1 = 0$$

$$\therefore D^2 = -1$$

$$\therefore D = \pm i$$

$$\text{C.F.} = C_1 \cos z + C_2 \sin z$$

Now, we use variation of parameter for finding P.I.

$$\therefore y_1 = \cos z \quad \text{and} \quad y_2 = \sin z$$

$$W = \begin{vmatrix} \cos z & \sin z \\ -\sin z & \cos z \end{vmatrix} = \cos^2 z + \sin^2 z = 1$$

$$W_1 = -y_2 = -\sin z$$

$$W_2 = y_1 = \cos z$$

Now,

$$\text{P.I.} = y_1 \int \frac{W_1}{W} r(x) dx + y_2 \int \frac{W_2}{W} r(x) dx$$

where,

$$r(x) = z \sin z$$

$\therefore$

$$\text{P.I.} = \cos z \int \frac{-\sin z}{1} \cdot z \sin z dz + \sin z \int \frac{\cos z}{1} \cdot z \sin z dz$$

$$= -\cos z \int z \cdot \sin 2z dz + \frac{\sin z}{2} \int \sin 2z \cdot z dz$$

$$= -\cos z \int z \cdot \left(\frac{1 - \cos 2z}{2} \right) dz + \frac{\sin z}{2} \int z \sin 2z dz$$

$$= -\cos z \left[\int \frac{z}{2} dz - \frac{1}{2} \int z \cdot \cos 2z \right] + \frac{\sin z}{2} \int z \cdot \sin 2z dz$$

Now

$$\int z \cos 2z dz = z \frac{\sin 2z}{2} - \left(\frac{-\cos 2z}{4} \right)$$

$$= \frac{z \sin 2z}{2} + \frac{\cos 2z}{4}$$

$$\int z \sin 2z dz = z \frac{\cos 2z}{2} - \left(\frac{\sin 2z}{4} \right)$$

$\therefore$

$$\text{P.I.} = -\cos z \left[\frac{z^2}{4} - \frac{1}{4} z \sin 2z - \frac{1}{8} \cos 2z \right] + \frac{\sin z}{2} \left[\frac{z \cos 2z}{2} - \frac{\sin 2z}{4} \right]$$

$$= -\frac{1}{4} z^2 \cos z + \frac{1}{4} z \cdot \cos z \cdot \sin 2z + \frac{1}{8} \cos z \cdot \cos 2z + \frac{1}{4} z \cdot \sin z \cdot \cos 2z - \frac{1}{8} \sin z \cdot \sin 2z$$

$\therefore$

$$y = \text{C.F.} + \text{P.I.}$$

Ex. 3.6.10 : Solve the differential equation $x^2 y'' - 2xy' + 2y = x^3 \cos x$.

GTU : Summer-19, Maths-2, Marks 7

Sol. : Here, $x^2 y'' - 2xy' + 2y = x^3 \cos x$ is given

Let, $x = e^z \Rightarrow z = \log x$

$$x \frac{dy}{dx} = Dy$$

$$x^2 \frac{d^2 y}{dx^2} = D(D-1)y \quad D = \frac{d}{dz}$$

$$(D(D-1) - 2D + 2)y = x^2 \cos x = e^{3z} \cos(e^z)$$

$$(D^2 - 3D + 2)y = e^{3z} \cos(e^z)$$

A.E.

$$D^2 - 3D + 2 = 0$$

$$(D - 2)(D - 1) = 0$$

$$D = 2, D = 1$$

∴

$$\text{C.F.} = y_C = C_1 e^{2z} + C_2 e^z$$

Compare with

$$\text{C.F.} = C_1 y_1 + C_2 y_2$$

$$y_1 = e^{2z} \quad \text{and} \quad y_2 = e^z$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{2z} & e^z \\ 2e^{2z} & e^z \end{vmatrix} = e^{3z} - 2e^{2z} = -e^z \neq 0$$

$$W_1 = -y_2 = -e^z$$

$$W_2 = y_1 = e^{2z}$$

$$\begin{aligned} y_P &= y_1 \int \frac{W_1}{W} r(x) dz + y_2 \int \frac{W_2}{W} r(z) dz \\ &= e^{2z} \int \frac{-e^z}{-e^z} e^{3z} \cos(e^z) dz + e^z \int \frac{e^{2z}}{-e^z} e^{3z} \cos(e^z) dz \\ &= e^{2z} \int e^{3z} \cos(e^z) dz \\ &= e^z \int e^{4z} \cos(e^z) dz \end{aligned}$$

$$\text{Now, } \int e^{3z} \cos(e^z) dz$$

Take

$$e^z = t \Rightarrow e^z dz = dt$$

$$\begin{aligned} \int (e^z)^2 \cos(e^z) e^z dz &= \int \underset{u}{t^2} \underset{v}{\cos t} dt = t^2(-\sin t) - (2t)(-\cos t) + 2(+\sin t) \\ &= \int e^{4z} \cos(e^z) dz = \int (e^z)^3 \cos(e^z) e^z dz \end{aligned}$$

Let,

$$e^z = t \Rightarrow e^z dz = dt$$

$$\begin{aligned} \int t^3 \cos t dt &= (t^3)(-\sin t) - (3t^2)(-\cos t) + (6t)(\sin t) - 6 \cos t \\ &= -t^3 \sin t + 3t^2 \cos t + 6t \sin t - 6 \cos t \end{aligned}$$

∴

$$\begin{aligned} y_P &= e^{2z}(-e^{2z} \sin(e^z) + 2e^z \cos(e^z) + 2 \sin(e^z)) \\ &\quad - e^z(-e^{3z} \sin(e^z) + 3e^{2z} \cos(e^z) + 6e^z \sin(e^z) - 6 \cos(e^z)) \\ &= -e^{4z} \sin(e^z) + 2e^{3z} \cos(e^z) + 2e^{2z} \sin(e^z) + e^{4z} \sin(e^z) \\ &\quad - 3e^{3z} \cos(e^z) - 6e^{2z} \sin(e^z) + 6e^z \cos(e^z) \\ &= -e^{3z} \cos(e^z) - 4e^{2z} \sin(e^z) + 6e^z \cos(e^z) \end{aligned}$$

$$y = y_C + y_P$$

$$= C_1 e^{2z} + C_2 e^z - e^{3z} \cos(e^z) - 4e^{2z} \sin(e^z) + 6e^z \cos(e^z)$$

$$y = C_1 x^2 + C_2 x - x^3 \cos x - 4x^2 \sin x + 6x \cos x$$

Ex. 3.6.11 : Solve the differential equation :

$$x^2 \frac{d^2 y}{dx^2} - 4x \frac{dy}{dx} + 6y = x^2 \log x \text{ using method of variation of parameters.}$$

Sol. : This is a Cauchy's linear equation with variable coefficients.

Putting $x = e^t \quad \therefore \log x = t$

$$\Rightarrow x \frac{dy}{dx} = Dy, \quad x^2 \frac{d^2 y}{dx^2} = D(D-1)y$$

$\therefore$ Given differential equation may be rewritten as

$$(D(D-1) - 4D + 6)y = te^{2t}$$

$$\Rightarrow (D^2 - 5D + 6)y = te^{2t}$$

Auxiliary equation is : $(m-2)(m-3) = 0$

$$\Rightarrow m = 2, 3$$

$$\text{C.F.} = C_1 e^{2t} + C_2 e^{3t} = C_1 y_1 + C_2 y_2$$

$$\therefore y_1 = e^{2t} \quad \text{and} \quad y_2 = e^{3t}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{2t} & e^{3t} \\ 2e^{2t} & 3e^{3t} \end{vmatrix} = e^{5t}$$

$$u_1 = -\int \frac{y_2 F(t)}{W} dt = -\int \frac{e^{3t} te^{2t}}{e^{5t}} dt = -\int t dt = -\frac{t^2}{2}$$

$$u_2 = \int \frac{y_1 F(t)}{W} dt = \int \frac{e^{2t} te^{2t}}{e^{5t}} dt = -\int te^{-t} dt = [(t)(-e^{-t}) - (1)(e^{-t})] = -te^{-t} - e^{-t}$$

$$\begin{aligned} \therefore \text{P.I.} &= u_1 y_1 + u_2 y_2 = -\frac{t^2}{2} e^{2t} - (te^{-t} + e^{-t}) e^{3t} \\ &= -\frac{t^2}{2} e^{2t} - te^{2t} - e^{2t} = -e^{2t} \left(\frac{t^2}{2} + t + 1 \right) \end{aligned}$$

Complete solution is :

$$y = \text{C.F.} + \text{P.I.}$$

$$\Rightarrow y = C_1 e^{2t} + C_2 e^{3t} - e^{2t} \left(\frac{t^2}{2} + t + 1 \right)$$

or
$$y = C_1 x^2 + C_2 x^3 - x^2 \left(\frac{(\log x)^2}{2} + \log x + 1 \right)$$

$$\Rightarrow y = C_3 x^2 + C_2 x^3 - \frac{x^2}{2} (\log x)^2 - x^2 \log x, \quad C_3 = C_1 - 1$$

